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Optimizing autonomous multimodal last-mile delivery systems with time windows: Analyzing trade-offs between drones, robots, and trucks

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A B S T R A C T

This paper studies multi-modal last-mile delivery systems. More specifically, it introduces the Drone-Assisted Vehicle Routing Problem with Robot Stations and Time Windows (VRPD-RS-TW) to face exigent delivery schedules in last-mile logistics. The VRPD-RS-TW applies to highly populated city centers, where due to infrastructural issues, customers might be hard to reach by conventional trucks. This problem is characterized by utilizing three transportation modes: trucks, drones, and robots collaborating to deliver parcels and meet time window requirements. To solve this problem, we develop a Mixed Integer Linear Programming (MILP) formulation, valid inequalities to strengthen the linear relaxation, and a Three-Phased Granular Multi-Start Iterated Local Search (3P-GMS-ILS) metaheuristic algorithm. A computational study on instances of Amsterdam (the Netherlands) demonstrates that the 3P-GMS-ILS outperforms the MILP formulation and state-of-the-art metaheuristic approaches in terms of both objective function values and computational times. Furthermore, we provide insights into the effectiveness of the VRPD-RS-TW, showing that, under various cost scenarios, the combination of trucks, drones, and robots outperforms scenarios with fewer transportation modes. Across the studied cost scenarios, the VRPD-RS-TW simultaneously coordinates a higher volume of delivery operations, with 48.75 % performed by drones and robots, resulting in better objective function values than the related transportation systems. In scenarios where truck routing costs are lower than those of drones and robots, the VRPD-RS-TW still achieves better objective function values compared to the related transportation systems, although the share of drone and robot deliveries drops to 14.78 %. This shows that adding drones is not necessarily beneficial when trucks carry drones, as trucks should wait for drones, which might result in higher costs related to drone routing and drivers' wages.

1. Introduction

Multimodal transport has shown substantial benefits along the whole transportation chain, i.e., pre-haul, long-haul, and end-haul (SteadieSeifi et al., 2014). Furthermore, by incorporating different transportation modes, such as trucks, trains, and ships, companies are able to exploit the benefits of different transport modes to improve the efficiency, reliability, and sustainability of the supply chain (Kelle et al., 2019).

This paper focuses on last-mile delivery in urban environments, which are characterized by the interaction of three main stakeholders: customers that search for a cost-reasonable, fast, and reliable delivery service, companies that place their interests in reducing

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costs, and governments that regulate the physical space in which the last-mile operations occur (Kiba-Janiak et al., 2021). Over the last few years, the rapid surge of online product sales and home deliveries has caused an increment in transportation externalities such as accidents, noise pollution, and gas emissions (Digiesi et al., 2017), which has increased the complexity of last-mile logistics operations, encompassing up to 30 % of the e-logistic costs (Wang et al., 2014). This has resulted in governmental authorities pushing companies to reduce the externalities by imposing restrictive access regulations that force them to develop new technologies and sustainable delivery systems (Dayarian et al., 2020). Last-mile operations should cope with unfavorable and complex environments, where the transfer of goods has become inefficient and expensive due to customers that present specific requirements and conditions, competition among companies, availability of parking areas, access to curbside, access to buildings/stores/houses, and readiness of the receivers (Jaller et al., 2020). Incorporating autonomous vehicles into last-mile delivery operations, e.g., drones and robots, appears as a practical approach to deal with transportation externalities while providing significant versatility to exploit the opportunities that multimodality offers (Campuzano, 2024). Nonetheless, each mode of transport provides its pros and cons regarding load capacity, battery endurance, travel speed, and transportation costs, among others (Gao et al., 2023). For this reason, extensive research and investments are being globally made to combine and incorporate the strengths of each modality in last-mile delivery (Zhang and Kamargianni, 2022), e.g., combinations of trucks with electric vehicles, drones, cargo bikes, parcel lockers, and autonomous robots.

Nowadays, companies can simultaneously coordinate numerous delivery operations by deploying autonomous drones that help reduce operational costs and improve service quality. This resulted in increasing interest in drones from the commercial sector, exemplified by promising projects, such as Parcelcopter by DHL, Wings by Google, and Air Prime by Amazon. Autonomous drones have also drawn the interest of the transportation research community, where scholars have devoted their efforts to developing combinations of multiple modes of transport that exploit the advantages of drones, e.g., avoidance of traffic congestion and infrastructural issues, high speeds, flying mobility, and reduced costs (Mulumba and Diabat, 2024). Accordingly, new research gaps have been identified where drones can be employed in various civil applications, including defense, territory monitoring, transportation and logistics, disaster relief, and farming (Masmoudi et al., 2022). Nevertheless, these applications raise new challenges regarding drone limitations and requirements that might lead to unfeasible solutions, for instance, battery life, deliveries that exceed the drone's capacity, no-flying zone restrictions, unsafe landing territory, vulnerability to difficult weather conditions, and mandatory signature for receiving packages. For this reason, drones are mainly involved in applications that require ground vehicles or trucks to support drone delivery operations, such as the Flying Sidekick Traveling Salesman Problem (Murray and Chu, 2015) and the Traveling Salesman Problem with Drone (Roberti and Ruthmair, 2021).

Autonomous robots are another transportation mode considered in the logistics sector. These robots are mainly applied to large warehouse problems, where robots have shown successful implementations (Sun et al., 2021). Autonomous robots do not face labor constraints and are equipped with multiple sensors and navigation technologies that allow them to travel on roads and sidewalks. This brings more opportunities and reduces transportation times for hard-to-reach customers (Yu et al., 2020). The use of robots in last-mile logistics has followed two main trends: (i) van-based delivery systems, in which one or several vans carry one or multiple robots to deploy them in parking sites and perform delivery operations (Jennings and Figliozzi, 2020), and (ii) micro hub-based systems, in which multiple robots are located in decentralized depots that are supplied with parcels by either vans or trucks to serve customers (Poeting et al., 2019b). Furthermore, robots are safer than drones since they have to drive at walking speeds for safety issues and governmental regulations. This has drawn the attention of companies, such as Amazon, with the Scout Sidewalk Robot project, and Starship Technologies, with the Starship Robot, which have developed robot-related delivery systems. In such systems, the key features identified are meeting specific time-window constraints and allowing customers to pick up their orders directly from the robots on the sidewalk (Poeting et al., 2019b).

Using multimodality to simultaneously coordinate delivery operations is a practical approach when facing time window constraints in last-mile logistics (Li et al., 2020; Chen et al., 2021; Kuo et al., 2022). These constraints are intrinsically present in the last mile and represent specific working conditions, customer requirements, and delivery restrictions, such as governmental policies, customer availability, perishable products' endurance, and service deadlines. Therefore, time windows are helpful in modeling realistic scenarios, appearing as either a deadline or a time interval at which a given parcel should be delivered. These constraints are classified as hard if they cannot be violated and soft when a specific penalty is incorporated into the objective function per each violation (Calvete et al., 2004).

In this paper, we propose the Drone-Assisted Vehicle Routing Problem with Robot Stations and Time Windows (VRPD-RS-TW). The VRPD-RS-TW depicts a novel delivery system incorporating multiple modes of transport into the delivery operations to improve the delivery system's efficiency, reduce operational costs, and meet time window requirements. Therefore, the main focus of this paper is to provide insights into the effectiveness of multi-modality in the last mile and study under which scenarios the decision-maker can obtain the highest benefits. Our contributions are summarized as follows:

(1) We present a novel extension of the VRPD-RS, denoted as the Drone-Assisted Vehicle Routing Problem with Robot Stations and Time Windows (VRPD-RS-TW). This variant investigates the collaborative efforts of trucks, drones, and robots in addressing realistic scenarios related to customer time windows. Additionally, we extend the VRPD-RS by considering extra elements in drone delivery operations, including features such as no-fly zones, and the impact of parcel weight and hovering time on the drone energy consumption. We formulate a Mixed-Integer Linear Programming (MILP) model to solve the VRPD-RS-TW and enhance its linear relaxation by introducing several valid inequalities.
(2) We develop a Three-Phased Granular Multi-Start Iterated Local Search to solve the VRPD-RS-TW. The 3P-GMS-ILS integrates granular neighborhoods with a multi-start mechanism to adjust the set of arcs that are explored within the neighborhoods and guide the search toward promising regions of the solution space. Moreover, due to the complexity of the delivery activities of

each transport modality, we provide a detailed description of how to compute the arrival, service, release, and departure times of trucks, drones, and robots to synchronize their delivery activities.

(3) We provide insights into the effectiveness of multimodality in last-mile logistics under different operational cost objective functions. We do this by analyzing how the VRPD-RS-TW utilizes the different modalities to minimize operational costs and the makespan. Furthermore, we analyze the trade-offs between the makespan and operational costs objective functions, providing insights into the complexity of these objectives and the potential benefits for the companies. In addition, considering different cost settings, we compare the performance of VRPD-RS-TW with other delivery concepts, i.e., (*i*) only-truck (*ii*) truck-and-drone, and (*iii*) truck-and-robot delivery systems. Finally, we present a sensitivity analysis on the key parameters that balance diversification and intensification in the 3P-GMS-ILS.

The remainder of this paper is organized as follows. A literature review is presented in Section 2. Next, we formally introduce and define the VRPD-RS-TW in Section 3. Then, Section 4 describes our heuristic solution methods. Section 5 contains the numerical experiments, parameter tuning, and results. After that, Section 6 examines the paper's underlying assumptions and summarizes key insights. Finally, we conclude with remarks and future research directions in Section 7.

2. Literature review

This literature review encompasses recent studies on last-mile multimodal transportation problems that incorporate autonomous vehicles, addressing truck-and-drone-and-robot-related problems comparable to the VRPD-RS-TW. For these problems, we highlight the applications, approaches, and results of last-mile delivery studies that lie within the scope of our interest. Section 2.1 studies truck and drone problems, Section 2.2 focuses on truck and robot problems, and Section 2.3 summarizes the literature on problems that combine trucks, drones, and robots in last-mile logistics.

2.1. Truck and drone delivery systems

Over the last few years, providers of last-mile delivery services have studied the use of drones. This has generated a vast body of literature devoted to this topic, where problems such as the Flying Sidekick Traveling Salesman Problem (FSTSP; Murray and Chu, 2015) and the Traveling Salesman Problem with Drone (TSP-D; Agatz et al., 2018) have given rise to a new research stream focusing on versatile truck-and-drone delivery systems. In the last mile, truck-and-drone delivery systems aim at providing solutions that reduce operational costs and transportation times, improve the customer experience, and deal with governmental regulations, e.g., urbanization policies that incite the development of environmentally friendly and sustainable industrial activities. However, these systems are more complex than the classic truck-only systems since multimodality involves synchronizing the delivery activities of trucks and drones (Moshref-Javadi et al., 2020). This section centers on the Vehicle Routing Problem with Drones (VRPD; Wang et al., 2017), which can be seen as a generalization of the FSTSP or TSP-D. Accordingly, we briefly summarize the VRPD-related literature, emphasizing the solution methodologies and problem features. For more extensive literature reviews on drone optimization problems and applications, the reader is referred to Coutinho et al. (2018) and Otto et al. (2018).

Several problem variants that exploit the advantages of drone-assisted transportation systems have been proposed in the recent literature. As such, further complexity is incorporated into the VRPD when considering the *en-route* operations (VRPDERO; Schermer et al., 2019), where trucks are allowed to retrieve and launch drones at specific locations while traversing the arcs. Das et al. (2020) develop a Pareto Ant Colony Optimization (P-ACO) algorithm to solve a bi-objective transportation system that minimizes travel distance and maximizes customer satisfaction. Raj and Murray (2020) extends the multiple FTSP (mFTSP) to introduce the Multiple Flying Sidekick Traveling Salesman Problem with Variable Drone Speeds (mFSTPS-VDS). The mFSTPS-VDS treats the drone speed as a decision variable to reduce the makespan and improve the drone utilization rate while respecting energy constraints. Bakir and Tiniç (2020) propose the Vehicle Routing Problem with Flexible Drones (VRPFD), where drones can assist multiple trucks and trucks can visit multiple times a given node to facilitate truck-drone synchronizations. The VRP with Same Day Delivery and Drone Resupply (VRPDR) is introduced in Dayarian et al. (2020). Here, drones resupply trucks with orders placed while trucks are en route. Di Puglia Pugliese et al. (2020) extend the VRPDTW from Pugliese and Guerriero (2017) to propose the Drone Constrained VRPDTW (DcVRP-DTW) and the Drone Constrained VRPD with Partial Time Windows (DcVRP-DpTW). In these problems, soft time windows are considered for the customers served by drones. Dayarian et al. (2020) study the Vehicle Routing Problem with Drone Resupply (VRP-DR) to test restricted and flexible resupply approaches. Campuzano et al. (2021) develop a multi-start variable neighborhood search metaheuristic for the TSP-D with energy constraints. Boccia et al. (2021) propose a MILP formulation for a variant of the FTSP, which allows trucks to wait at the launching locations for drones. The MIlP formulation is an extension of the FTSP model presented in Murray and Chu (2015), which avoids drone variables with 3 sub-indexes. Boccia et al. (2021) implement a branch-and-cut algorithm, solving to optimality instances of up to 20 customers. Coindreau et al. (2021) study the Minimum Cost Vehicle Routing Problem with Time Windows and Drones (MC-VRPTW-D) to minimize operational costs involving truck fuel costs, driver wages, and drone electricity costs. Campuzano et al. (2023) introduce the Variable Speeds Asymmetric Traveling Salesman Problem with Drone (VS-ATSP-D), where the asymmetric drone transportation times depend on the weather conditions and parcel load. Di Puglia Pugliese et al. (2021) study the VRPD-TW considering a time-dependent energy consumption function of drones. Momeni et al. (2022) propose a multi-objective variant of the VRPD for bushfire prevention and detection to monitor impassable areas for trucks. Kloster et al. (2022) introduce the Multiple Traveling Salesman Problem with Drone Stations (mTSP-DS), where drones are deployed as round-trips. El-Adle et al. (2023) develop a variable neighborhood search (VNS) algorithm to study the TSP-D with round trips only. The VNS algorithm uses a two stages intensification scheme and a MILP formulation to solve carriers' routes. The results show improvements of up to 1 %–8 % in delivery times when multiple round trips are allowed. Boschetti and Novellani (2023) introduce the traveling salesman problem with drone and lockers (TSP-DL), adding an additional dimension to the delivery system where lockers, served by trucks, can be used for customers to pick up the parcels. Boschetti and Novellani (2023) propose a set of MILP formulations and a branch-and-cut algorithm to solve the TSP-DL. The results show that lockers allow more flexibility in the delivery system, generating reductions in costs. Ma et al. (2024) study the Two-Echelon Drone and Electric Vehicle Joint Delivery Route Planning Problem (2E-VRPD), which aims to minimize both economic and environmental costs. To solve the 2E-VRPD, the authors propose a hybrid genetic algorithm enhanced with large-scale neighborhood search. The results indicate that increasing the drone endurance radius leads to a consistent reduction in total delivery costs.

2.2. Truck and robot delivery systems

Incorporating autonomous robots into logistics operations has resulted in implementations in warehousing environments (Sun et al., 2021). Robots have also been used outside warehousing environments for last-mile delivery operations. These vehicles present several advantages to trucks and drones. As opposed to drones, they do not face restricted flying zones, have a bigger storage capacity, but might be able to carry bigger batteries (Ostermeier et al., 2021). On the other hand, in comparison to trucks, autonomous robots drive at slower speeds, and face other forms of congestion as they drive on pedestrian streets, but might be able to avoid infrastructure issues generated, for instance, by traffic lights (Ostermeier et al., 2021). These features have uncovered many opportunities for autonomous robots in last-mile logistics, generating an increasing research interest in recent years.

Special attention has been given to problems where trucks carry robots as a mobile depot and deploy them to perform deliveries. The Robot-Assisted Truck Delivery Problem (TSP-R) is a particular case of the TSP-D, where robots are dispatched from trucks and can perform more than one delivery (Simoni et al., 2020). To solve the TSP-R, the authors develop an integer linear programming (ILP) formulation that uses precompiled operations to minimize transportation times, and an iterated local search procedure with adaptive perturbations (LS-AP) that use dynamic programming to solve robots' routing sub-problems. Results show that the LS-AP achieves effective solutions in short computational times and that the robot delivery approach is useful in urban areas with dense population distribution and high levels of traffic congestion. Besides, results point out that the maximum driving range and the drop-off time are important features for the efficiency of the delivery system, and that splitting the storage capacity improves the efficiency of the robots. Yu et al. (2020) consider a two-echelon version of the VRPR (2E-VRPR), where trucks (or large vehicles) carry autonomous robots in the first level route, and drop off and pick up robots in rendezvous nodes. Then, the robots should perform the deliveries in the second level route and meet time window constraints. They propose a MILP formulation and a column generation (CG) algorithm to improve the MILP lower bounds. Furthermore, a metaheuristic procedure incorporating operations such as multi-start, destroy, repair, iterative local search, and backtracking is proposed. In the experiments, a sensitivity analysis shows that increasing autonomous vehicles' speed has a limited effect on cost. Chen et al. (2021) consider a fleet of vans equipped with several self-driving robots to serve the customers while minimizing transportation times and meeting time windows constraints. They call this problem the Vehicle Routing Problem with Time Windows and Delivery Robots (VRPTWDR), provide a MILP formulation, and develop an ALNS. In their findings, the ALNS shows high-quality solutions in instances of up to 200 customers in reduced computational times. Furthermore, results demonstrate that the implementation of self-driving robots provides effective solutions to delivery environments characterized by densely populated areas with a limited number of parking locations. Yu et al. (2022) introduce the two-echelon electric van-based robot delivery system with *en-route* charging (2E-VREC). The en-route charging approach allows robots to charge their batteries while vans carry them. The authors develop a MILP formulation and an ALNS algorithm. They conduct a sensitivity analysis, finding that en-route charging reduces transportation costs compared to the static-charging approach when studying different time window lengths, charging rates, and battery capacity settings.

Another research field focuses on delivery systems where one or more trucks carry all the packages and use micro hubs equipped with autonomous robots as transfer points to perform the deliveries. A case study is performed by Poeting et al. (2019a), where a two-tier time-window delivery system consisting of a single truck and multiple robots (2T-TSP-R-TW) is studied in the city of Cologne, Germany. They develop a heuristic algorithm, which in a constructive phase solves a traveling-salesman MILP, and then applies a simulated annealing heuristic (SA) for the allocation and scheduling of robots. Results show that in the two-tier approach, the locations of the micro hubs are crucial. The Uncapacitated Routing-Scheduling Problem (URSP) is a problem where a single truck serves several facilities, which deploy delivery robots that perform the last-mile deliveries (Alfandari et al., 2021). The URSP focuses on meeting customers' deadlines, incorporating penalty costs in the objective function. The authors study three different variants for this problem, propose a MILP formulation, and develop a Bender's decomposition in combination with a branch-and-cut scheme. Results show that the branch-and-Benders-cut method can efficiently solve the URSP for large instances. Bakach et al. (2021) propose a p-median-based mathematical formulation for the 2T-TSP-R-TW. They solve instances of up to 300 customers, showing that the two-tier robot-based delivery approach can reduce conventional delivery costs by 24–32 %. Campuzano et al. (2025) introduce the Two-Tier Multi-Depot Vehicle Routing Problem with Robot Stations and Time Windows (2T-MDVRP-RS-TW), in which the delivery process is split into two tiers: trucks transport parcels from multiple depots to robot stations, and from there, sidewalk robots perform round-trip deliveries to customers. To solve the 2T-MDVRP-RS-TW, the authors develop a multi-start iterated local search metaheuristic with crossover and repair operators (MS-ILS-CR). Results highlight the significant impact of depot configuration, showing that increasing the number of depots from one to three can reduce operational costs by up to 56 %.

Some studies also combine different approaches with trucks carrying robots and robots available in micro hubs or robot stations. Boysen et al. (2018) study the Truck-Based Robot Delivery Scheduling (TBRD) problem, where one truck loaded with delivery robots can use decentralized robot depots to serve a set of customers. The objective of the TBRD is to minimize the penalty costs related to customers' deadlines. A MILP formulation and a multi-start local search algorithm (MLSA) are proposed. Here, the mathematical formulation is only able to provide good-quality solutions for a small set of instances. In contrast, the MLSA finds competitive solutions for small and large instances. Additionally, the results reveal that the delivery system can substantially benefit from implementing autonomous robots, which help reduce the truck fleet. Sonneberg et al. (2019) propose an optimization model to minimize the operational costs for the Electric Location Routing Problem with Time Windows (ELRP-TW). The ELRP-TW searches for a number of locations to be opened, the assignment of customers, and the driving routes while meeting constraints such as battery capacity and time windows. In the experiments, the authors study the effectiveness of providing autonomous vehicles with multiple compartments, finding that a second compartment reduces the total costs by nearly 46 %. Ostermeier et al. (2021) study a truck and robot problem where trucks carry packages along with robots to meet time windows (VRPR-TW). As such, trucks can park either at a robot depot, which has additional robots, or at a parking location to deploy robots and serve customers. The authors propose a MILP model and develop two three-step heuristics, denoted by truck-and-robot cost-optimal routing approach (TRC) and TRC robot scheduling heuristic (TRC-RSH). The latter approach uses a tailored heuristic based on a local search algorithm to improve truck routing. The authors show that truck distances and emissions are reduced by more than 60 %, demonstrating that the truck and robot approach outperforms the classic truck delivery system in terms of cost and service criteria. Heimfarth et al. (2022) propose the Mixed Truck and Robot Routing Problem (MTR-RP), where trucks are allowed to serve customers, pick up robots at robot depots, and deploy robots at specific drop-off locations for delivery. After delivering a parcel, robots should return to the closest robot depot. This way, the MTR-RP searches for minimizing the operational costs while meeting time windows constraints. The authors develop a MILP formulation and a General Variable Neighborhood Search (GVNS) to solve the MTR-RP. Results show that the GVNS outperforms existing algorithms by at least 37 %, and costs are being reduced by more than 40 % compared to truck-only deliveries.

2.3. Truck, drone, and robot delivery systems

As discussed in Sections 2.1 and 2.2, the deployment of autonomous vehicles, such as drones and robots, in delivery activities can mitigate the effect of the last-mile transportation externalities (Lemardelé et al., 2021). Accordingly, this section focuses on studying problems that exploit the potential combinations of these technologies in last-mile logistics. This section finishes with a discussion of the research gap this paper aims to fill. The reader is referred to Jaller et al. (2020) for a more detailed explanation of the opportunities and challenges of autonomous technologies in last-mile logistics.

Moeini and Salewski (2019) introduce one of the first autonomous multimodal problems in the literature, where one truck carries drones and autonomous transport vehicles (ATVs) to deliver customers' parcels. The truck is allowed to travel among grid points (parking sites) to dispatch the vehicles that perform the deliveries while minimizing the makespan. The problem is called the Truck-Drone-ATV Routing Problem (TDA-RP), and a genetic algorithm (GA) incorporating a Lin-Kernighan heuristic to create a TSP route for the truck is developed. The GA finds solutions for instances of up to 75 customers, and results demonstrate that implementing drones and self-driving robots helps reduce the final makespan. A different study focused on mitigating traffic congestion issues in food delivery is performed by Samouh et al. (2020). The authors compare the effects of drones, robots, and hybrid drone-and-robot delivery systems in the last mile. An agent-based simulation is carried out on a case study considering the city of Mississauga, Canada. Results show that the hybrid delivery approach outperforms the independent drone and robot systems, reducing the preparation and delivery time by 48 % and 42 % for each problem, respectively. The Drone-Assisted Traveling Salesman Problem with Robot Stations (TSP-D-RS), introduced by Schermer et al. (2020), is a multimodal problem in which a single truck assisted by a drone is allowed to activate robot stations equipped with autonomous robots to serve customers. A MILP formulation that finds optimal solutions for instances up to 15 nodes is proposed. Results show that minimizing the makespan reduces operational costs. However, minimizing the operational costs might generate an increment in the makespan. Morim et al. (2024) extend the TSP-D-RS from Schermer et al. (2020) and propose the Drone-Assisted VRP with Robot Stations (VRPD-RS). The VRPD-RS is a problem in which trucks, in collaboration with drones, can use robot stations to serve customers with their corresponding parcels. To solve this problem, the authors introduce a MILP formulation and develop a general variable neighborhood search algorithm (GVNS) consisting of three phases: building a VRP solution, inserting drone operations, and incorporating robot operations. In their experiments, the makespan and operational costs objective functions are studied. Their results show that the MILP is unable to optimally solve instance sets larger than 10 customers for the makespan objective function, while optimal solutions are found for instance sets of up to 20 customers for the costs objective function.

In particular, our research extends the VRPD-RS from Morim et al. (2024) by introducing the VRPD-RS-TW. While the two models share similarities, the VRPD-RS-TW removes several simplifying assumptions, thereby increasing both the modeling and algorithmic complexity of the optimization approaches. First, we incorporate energy constraints into drone operations by discretizing the energy consumption on each arc and accounting for the impact of parcel weight. Second, we eliminate the assumption that drones can land and wait for trucks on the ground, as this could result in unintended damage. To address this, we convert energy constraints into time-based constraints, thereby limiting the maximum flight time, including hovering times when required (see Section 5.1.4). Third, we introduce time-window constraints into the modeling framework, recognizing that autonomous vehicles and last-mile delivery operations often require attended deliveries. Although this may seem straightforward, it introduces substantial modeling challenges, as it becomes necessary to coordinate the sequence of delivery activities to ensure that customer time windows are met. As a result, the MILP formulation grows considerably more complex, and novel neighborhood structures are required to explore delivery operation combinations that are entirely neglected in the VRPD-RS. Mokhtari-Moghadam et al. (2025) introduce a Multi-Objective Vehicle Routing Problem with Drones and Robots (MOVRP-DR), aiming to minimize delivery costs while maximizing

Table 1
Overview of the literature related to the VRPD-RS-TW.

Reference	Acronym	Trucks	Drones	Robots	Time windows	Drone's range	Truck's capacity
Wang et al. (2017)	VRPD	n	m			distance	✓
Pugliese and Guerriero (2017)	VRPDTW	n	m		✓	distance	✓
Boysen et al. (2018)	TBDR	1		r	✓		
Schermer et al. (2018)	VRPD	n	m			time	✓
Schermer et al. (2019b)	VRPD	n	m			distance	✓
Sacramento et al. (2019)	VRPD	n	m			time	✓
Poeting et al. (2019a)	2T-TSP-R-TW	1		r	✓		
Wang and Sheu (2019)	VRPD	n	m			time	✓
Schermer et al. (2019)	VRPDERO	n	m			distance	
Moeini and Salewski (2019)	TDA-RP	1	m	r		time	
Sonneberg et al. (2019)	ELRP-TW			r	✓		✓
Schermer et al. (2020)	TSP-D-RS	1	1	r		time	
Raj and Murray (2020)	mFSTSP-VDS	n	m			energy	
Bakir and Tiniç (2020)	VRPFD	n	m			time	
Simoni et al. (2020)	TSP-R	1		r			
Yu et al. (2020)	2E-VRPR	n		r	✓	time	✓
Di Puglia Pugliese et al. (2020)	DcVRP-TW	n	m		✓	distance	✓
Dayarian et al. (2020)	VRP-DR	n	m		✓		
Alfandari et al. (2021)	URSP	1		r	✓		
Campuzano et al. (2021)	TSP-D	1	1			energy	
Coindreau et al. (2021)	MC-VRPTW-D	n	m		✓	time	
Tamke and Buscher (2021)	VRPD	n	m			distance	
Euchi and Sadok et al. (2021)	VRPD	n	m			time	✓
Chen et al. (2021)	VRPTWDR	n		r	✓		✓
Bakach et al. (2021)	2T-TSP-R-TW	1		r	✓		
Ostermeier et al. (2021)	VRPDR-TW	n		r	✓		✓
Di Puglia Pugliese et al. (2021)	VRPD-TW	n	m		✓	energy	✓
Lei et al. (2022)	VRPD	n	m			time	✓
Yu et al. (2022)	2E-VREC	n		r	✓		✓
Masmoudi et al. (2022)	VRPD-MC	n	m		✓	energy	✓
Gu et al. (2022)	VRPD-MV	n	m			energy	✓
Momeni et al. (2022)	VRPD	n	m			energy	
Kloster et al. (2022)	mTSP-DS	n	m			distance	
Heimfarth et al. (2022)	MTR-RP	n		r	✓		✓
Campuzano et al. (2023)	VS-ATSP-D	1	1			energy	
Ma et al. (2024)	2E-VRPD	n	m	r	✓	energy	✓
Morim et al. (2024)	VRPD-RS	n	m	r		time	✓
Campuzano et al. (2025)	2T-MDVRP-RS-TW	n		r	✓		✓
Chen et al. (2025)	VRP-HAA	n	m	r		time	✓
Mokhtari-Moghadam et al. (2025)	MOVRP-DR	n	m	r	soft	energy	✓
This work	**VRPD-RS-TW**	**n**	**m**	**r**	✓	**time, energy**	✓

customer satisfaction. They propose an adaptive multi-objective genetic algorithm (AMOGA) and benchmark it against NSGA-II, observing longer computational times but improved performance of AMOGA for half of the test instances. The results suggest that MOVRP-DR is particularly beneficial in scenarios with a moderate mix of high-density and remote customer locations. In particular, one of the main differences between this work and the VRPD-RS-TW model is that our approach captures more realistic last-mile delivery operations. Specifically, we consider: (*ii*) the effect of parcel weight on drone energy consumption, (*ii*) driver wage costs, (*iii*) hard time windows, (*iv*) no-fly zones, and (*v*) robots dispatched from dedicated robot stations rather than assisting trucks. Chen et al. (2025) introduce the Vehicle Routing Problem with Heterogeneous and Autonomous Assistants (VRP-HAA) to minimize the makespan, where no time windows are considered. To solve the VRP-HAA, they develop a hybrid metaheuristic that combines a genetic algorithm with large neighborhood search. They propose several valid inequalities to enhance the computational efficiency of the MILP formulation. Results show that deploying a single drone can reduce the makespan by up to 23.57 %, while using a single robot yields an improvement of up to 37.19 %.

Table 1 summarizes the relevant literature on both truck-and-drone and truck-and-robot problems, and presents their corresponding references. The second column of the table provides the problem acronyms. Furthermore, 'n', 'm', and 'r' report whether each modality is treated as a single unit or multiple units in each study. The sixth column indicates whether time window constraints are considered. The seventh column points out which type of constraint is used to restrict the drone flying range. Finally, the last column shows whether the problem restricts the truck's capacity.

As shown in Table 1, there is an increasing body of literature on autonomous vehicles in last-mile logistics. Most studies focus on minimizing either the makespan or operational costs. A subset of this work incorporates penalty costs into the objective function (Boysen et al., 2018; Poeting et al., 2019a; Heimfarth et al., 2022), while Ma et al. (2024) employs a convolutional neural network to estimate and minimize carbon emission costs. Notably, Masmoudi et al. (2022) frame the objective functions as profit maximization, optimizing delivery revenues minus routing costs, while restricting trucks by maximum working time.

Mokhtari-Moghadam et al. (2025) conduct a bi-objective study that simultaneously minimizes delivery costs and maximizes customer satisfaction, measured as expectation fulfillment, with satisfaction equal to 1 if the delivery occurs within the time window and 0 otherwise. Gu et al. (2022), Momeni et al. (2022), Masmoudi et al. (2022), Chen et al. (2025) study delivery systems where drones can carry several packages per sortie. Schermer et al. (2019) introduce a VRPD variant with en-route operations, enabling drones to be launched from and retrieved at discrete points located on the arcs traversed by trucks. Yu et al. (2022) propose an electric-van-based delivery system with dedicated parking nodes where vans both replenish their batteries and recharge on-board delivery robots. Campuzano et al. (2023) are the only authors to examine how weather conditions, parcel weight, energy consumption, and variable drone speeds jointly affect asymmetric drone transport times. Chen et al. (2025) uniquely study a VRPD-R delivery system featuring both heterogeneous drones and robots. Furthermore, most studies assume hard time windows, whereas only a few address soft time windows or due-date deliveries (Boysen et al., 2018; Poeting et al., 2019a; Di Puglia Pugliese et al., 2020; Dayarian et al., 2020; Alfandari et al., 2021; Heimfarth et al., 2022; Mokhtari-Moghadam et al., 2025).

In the studied literature, just a small stream of research considers the combination of multiple autonomous modes of transport (Moeini and Salewski, 2019; Schermer et al., 2020; Morim et al., 2024; Ma et al., 2024; Mokhtari-Moghadam et al., 2025), i.e., delivery systems that involve trucks, drones, and delivery robots. In this paper, we introduce the Drone-Assisted Vehicle Routing Problem with Robot Stations and Time Windows. The VRPD-RS-TW bridges a critical research gap regarding the combination of features, such as the effect of the parcel weight and hovering time on the drone energy consumption in the context of an autonomous multimodal delivery system. In addition, as shown in this literature review, there are several variants of (*i*) trucks-and-drones and (*ii*) trucks-and-robots delivery systems. However, a comprehensive examination of the advantages and limitations associated with each transportation modality remains elusive. Consequently, we execute a comparative analysis of these delivery systems with the VRPD-RS-TW. Our study aims to provide valuable insights into the benefits of each transport modality, particularly considering realistic urban scenarios for the city of Amsterdam, the Netherlands.

3. The drone assisted vehicle routing problem with robot stations and time windows

In this section, we formally introduce the VRPD-RS-TW. Section 3.1 provides a detailed description of the problem, its main aspects, and assumptions. The mathematical formulation and valid inequalities for the VRPD-RS-TW are presented in Section 3.2.

3.1. Problem description

The VRPD-RS-TW consists of finding a routing schedule in which trucks, drones, and delivery robots work in tandem to provide customers with their corresponding parcels. Here, delivery robots are located in micro hubs, also known as *robot stations*, and trucks carry drones. Robots serve customers in round-trips (same starting and ending locations), whereas drones can serve customers in the form of multi-legs (different take-off and landing locations) and round-trips. We refer to drone deliveries as *sorties*. Fig. 1 illustrates a feasible VRPD-RS-TW delivery schedule. In the example, two trucks and drones perform two independent routes to serve a given set of customers. The first route serves customers $\{1, 3, 4, 7, 10\}$ and activates robot station $\{13\}$. The second route visits customers $\{2, 5, 6, 8, 9, 11, 12\}$ and activates robot station $\{15\}$. Three different transportation modes are applied to serve customers $\{9, 11\}$ by trucks, customers $\{1, 3, 5, 12\}$ by drones, and customers $\{2, 4, 6, 7, 8, 10\}$ by robots. Depending on the type of vehicle that visits a node, the node is referred to as a *truck node*, *drone node*, or *robot node*. Nodes visited by trucks and drones are referred to as *combined nodes*. Moreover, two types of sorties are shown in the example, where the simultaneous coordination of the truck and drone is essential for a feasible VRPD-RS-TW implementation. For instance, in the multi-leg $\{D, 3, 13\}$, the truck must wait for the drone at the end of the sortie before heading to the depot, i.e., station $\{13\}$. Otherwise, the drone could not change the battery and load a new parcel to perform the next multi-leg $\{13, 1, D\}$. On the other hand, if we study the round-trip $\{11, 12, 11\}$, the truck must wait for the drone at the customer's location $\{11\}$ before departing to the depot. Note that the ending node of a drone sortie, where a drone meets the truck, can be either the depot, a customer location, or a robot station. We refer to this node as the meeting point. To model the VRPD-RS-TW, we make the following assumptions:

(a) Customers can be visited by trucks, drones, and robots. However, customers that belong to no-flying zones cannot be served by drones.
(b) There is a fleet of K trucks available at the depot. Each truck has a capacity Q and is assisted by a single drone.
(c) Trucks are allowed to arrive at a given location, park, and wait until the time window gets available to perform the delivery. However, drones and robots cannot park and wait.
(d) When finishing a given sortie, the drone can only land on the truck at the meeting point. Thus, if a drone arrives first at the meeting point, it should hover in the air until the truck's arrival.
(e) If a truck arrives first at the meeting point, the truck should wait for the drone's arrival before leaving the meeting point.
(f) Each robot station has R autonomous robots available to deliver parcels, where each robot can perform multiple round trips.
(g) Drones and robots should perform their delivery operations within the corresponding flying and driving times that are allowed by their batteries.
(h) Drones and robots can only carry one parcel per delivery operation.
(i) Robot stations cannot be used as parking sites to only deploy drones. Therefore, trucks can only activate robot stations if robots are deployed from them.
(j) Robot stations can be activated by only one truck. As such, packages can be delivered at stations and stay there until the time windows become available.

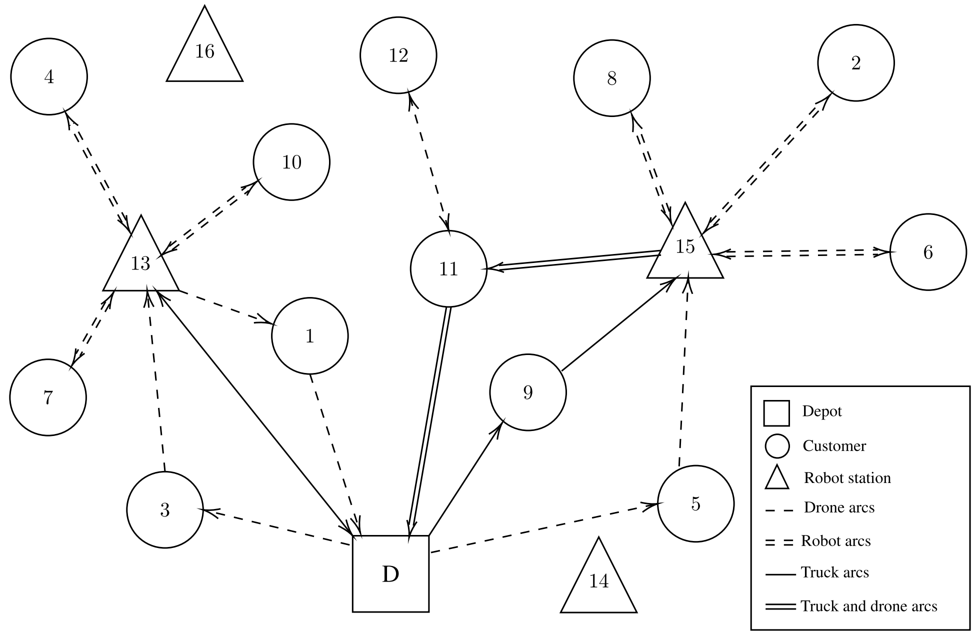


Fig. 1. Illustration of a VRPD-RS-TW delivery schedule.

(k) Constant values are considered for the parcel loading, unloading, delivery, taking-off, and landing times.

If the set of robot stations and the number of assisting drones are reduced to $\emptyset$, the VRPD-RS-TW turns into the Vehicle Routing Problem with Time Windows (VRP-TW), which is an $\mathcal{NP}$-hard problem (Savelsbergh, 1985; Li and Lim, 2003). Hence, the VRPD-RS-TW can be seen as a generalization of the VRP-TW, where any feasible route for the VRP-TW is a valid solution for the VRPD-RS-TW. This shows that the VRPD-RS-TW is at least as complex as the VRP-TW and, therefore, is an $\mathcal{NP}$-hard problem.

3.2. Mathematical formulation of the VRPD-RS-TW

This section provides the mathematical formulation of the VRPD-RS-TW. Even though the formulation is not intended for solving large-scale instances, the MILP serves as a benchmark and offers a rigorous basis for validating solution quality on small cases.

The VRPD-RS-TW is defined over a directed graph $\mathcal{G} = (V, A)$, with vertex set V and arc set A. The vertex set is defined as $V = \{0\} \cup V_{\mathcal{H}} \cup \{n+1\}$ and the arc set as $A = \left\{(0,j) | j \in V_{\mathcal{H}}\right\} \cup \left\{(i,j) | i,j \in V_{\mathcal{H}} : i \neq j\right\} \cup \left\{(i,n+1) | i \in V_{\mathcal{H}}\right\}$, where the vertices $\{0\}$ and $\{n+1\}$ represent the starting and ending points of the routes (the depot), $V_{\mathcal{N}} = \{1,\dots,n\}$ is the set of customers, $V_{\mathcal{S}} = \{1,\dots,s\}$ is the set of robot stations, and $V_{\mathcal{H}} = V_{\mathcal{N}} \cup V_{\mathcal{S}}$ groups all the nodes of the network that are different from the starting and ending depots $\{0\}$ and $\{n+1\}$. We refer to $V_{\mathcal{L}} = V \setminus \{n+1\}$ as the subset of vertices in V from which drones may be launched, $V_{\mathcal{R}} = V \setminus \{0\}$ as the subset of vertices where the drones may land, and $\mathcal{I}$ as the set of nodes located in the no-fly zones, where $\mathcal{I} \subset V_{\mathcal{N}}$.

The VRPD-RS-TW consists of $\mathcal{K} = \{1,\dots,K\}$ trucks that are initially located in the central depot, each one assisted by a single drone, which should perform delivery routes to serve customers $i \in V_{\mathcal{N}}$ with their corresponding parcels of weight q_i. The trucks' travel times to traverse an arc $(i,j) \in A$ are given by t^T_{ij}, with a service time of $\bar{l}^T$ at the customer locations and robot stations. In this delivery system, trucks and drones work in collaboration to serve customers, where drones are able to perform sorties in the form of multi-leg and round-trip delivery operations. In particular, the drone assisting a truck $k \in \mathcal{K}$ is allowed to perform multiple round-trips from a node $i \in V_{\mathcal{L}}$. This way, the pth round-trip from i is given by $p \in \mathcal{P} = \{0,..,P\}$, where $P = n-1$ represents the order in which the round-trips are performed from node $i \in V_{\mathcal{L}}$. The drones' travel times to traverse arc (i,j) are given by t^D_{ij}. Further, the drone's time to load a parcel is l^D and the time to deliver it is $\bar{l}^D$. Similarly, b^{on}_{ij} and b^{off}_{ij} represent the energy the drone consumes when traversing arc (i,j), for the case where the drone is flying with a parcel on- and off-board, respectively. Drones have a battery capacity of B, which is transformed into a maximum flying time L_{ijh}, to perform sorties that start from i serve customer j and finish at h, i.e., for arcs (i,j) and $(j,h) \in A$. Additionally, the robots' times to load a parcel are given by l^R and the time to deliver a parcel is $\bar{l}^R$. The VRPD-RS-TW allows every truck to activate robot stations. Each robot station has $\mathcal{R} = \{1,\dots,R\}$ autonomous robots available to drive through the city and deliver parcels. The time a robot spends to traverse arc (i,j) is given by t^R_{ij}, and its maximum driving time is T. The VRPD-RS-TW considers homogeneous fleets of trucks, drones, and robots, where trucks have a maximum load capacity of Q kilograms. The objective is to find valid routes that optimize operational costs ω, start and end at the depot for every truck, and visit every customer exactly once within the time-window intervals that start and finish at t^{start}_i and t^{stop}_i for customers $i \in V_{\mathcal{N}}$, respectively. The distances that trucks, drones, and robots cover to traverse arc $(i,j) \in A$ are given by $\widetilde{d}^T_{ij}$, $\widetilde{d}^D_{ij}$, and $\widetilde{d}^R_{ij}$, respectively. The routing costs per kilometer of trucks, drones, and robots are defined as c^T, c^D, and c^R, respectively. In addition to the routing costs, the

Table 2
Variables of the mathematical model.

ω	cost objective	$s^T_{ik} \in \mathbb{R}^+_0$ $\forall k \in \mathcal{K}, i \in V$	start of the service time of truck k at location i
$a^T_{ik} \in \mathbb{R}^+_0$ $\forall k \in \mathcal{K}, i \in V$	arrival time at i of truck k	$z_{ik} \in \mathbb{R}^+_0$ $\forall k \in \mathcal{K}, i \in V$	position of node i in the route of truck k
$a^D_{ipk} \in \mathbb{R}^+_0$ $\forall k \in \mathcal{K}, p \in \mathcal{P}, i \in V$	arrival time at i of the pth sortie, whose parcel comes from truck k	$x^T_{ijk} \in \{0,1\}$ $\forall k \in \mathcal{K}, i, j \in V$	equal to 1, if truck k traverses arc (i,j)
$d^T_{ik} \in \mathbb{R}^+_0$ $\forall k \in \mathcal{K}, i \in V$	departure time from i of truck k	$x^D_{\substack{ijh\\pk}} \in \{0,1\}$ $\forall k \in \mathcal{K}, p \in \mathcal{P}$ $i,h \in V, j \in V_{\mathcal{N}}$	equal to 1, if drone k performs the pth sortie from i to j to h
$d^D_{ipk} \in \mathbb{R}^+_0$ $\forall k \in \mathcal{K}, p \in \mathcal{P}, i \in V$	departure time from i of the pth sortie, whose parcel comes from truck k	$x^R_{\substack{ijk\\rp}} \in \{0,1\}$ $\forall k \in \mathcal{K}, p \in \mathcal{P}$ $r \in \mathcal{R}, i \in V_{\mathcal{S}}, j \in V_{\mathcal{N}}$	equal to 1, if robot r, whose parcel comes from truck k, serves customer j in a round trip from station i in the pth position
$r^R_{\substack{ir\\pk}} \in \mathbb{R}^+_0$ $\forall k \in \mathcal{K}, p \in \mathcal{P}$ $r \in \mathcal{R}, i \in V_{\mathcal{S}}$	release time of robot r from i to serve a customer in the pth robot delivery operation		

drivers' wages f^T are also incorporated into the operational costs objective function. The reader is referred to Section 5.1 for further details on the computation of operational costs, travel times, and energy consumption. The MILP formulation of the VRPD-RS-TW is given in (2)–(43), where the definition of the operational costs objective function ω is provided by Eq. (1). Note that Eq. (1) defines operational costs by aggregating the individual cost components, i.e., the arc costs and the driver costs proportional to total driving times (see Section 5.1.3). For the sake of compactness, all constraints that restrict the domains of binary variables $x^T_{ijk}, x^D_{\substack{ijh\\pk}}, x^R_{\substack{ijr\\pk}}$ to the set $\{0,1\}$ and continuous variables $a^T_{ik}, a^D_{ik}, d^T_{ik}, d^D_{ipk}, s^T_{ik}, r^R_{\substack{ir\\pk}}, z_{ik}$ to $\mathbb{R}^+_0$ are not listed explicitly. Table 2 provides the definition of the decision variables of the MILP.

$$\omega = \sum_{k\in K} f^T d^T_{n+1,k} + \sum_{k\in\mathcal{K}}\sum_{p\in\mathcal{P}}\sum_{i\in V_{\mathcal{L}}}\sum_{\substack{j\in V_{\mathcal{N}},\\ j\neq i}}\sum_{\substack{h\in V,\\ h\neq j}} c^D(\tilde{d}^D_{ij} + \tilde{d}^D_{jh})x^D_{\substack{ijh\\pk}}$$
$$+ \sum_{k\in\mathcal{K}}\sum_{i\in V_{\mathcal{L}}}\sum_{\substack{j\in V_{\mathcal{R}},\\ j\neq i}} c^T\tilde{d}^T_{ij}x^T_{ijk} + \sum_{k\in\mathcal{K}}\sum_{p\in\mathcal{P}}\sum_{r\in\mathcal{R}}\sum_{i\in V_{\mathcal{S}}}\sum_{j\in V_{\mathcal{N}}} c^R(\tilde{d}^R_{ij} + \tilde{d}^R_{ji})x^R_{\substack{ijr\\pk}} \tag{1}$$

The flow balance of the truck, drone, and robot delivery operations is guaranteed by Constraints (3)–(12). Constraints (3) restrict the number of times a truck k can leave the depot. Constraints (4) establish that every truck departing from the starting station $\{0\}$ should finish the route at the end station $\{n+1\}$. Constraints (5) indicate that every truck that arrives at a customer location must depart from it after the delivery. Constraints (6) ensure that every pth sortie is performed at most once. Constraints (7)–(8) indicate that drone sorties should start and finish at nodes that are visited by trucks. Constraints (9) establish that the $(p+1)$th sortie can only be performed if the previous pth sortie is already finished. Constraints (10) ensure that only one truck can visit a given robot station. Constraints (11) restrict the maximum number of robot delivery operations executed from a robot station. Constraints (12) establish that trucks can only visit a robot station if at least one sidewalk robot is deployed from the station.

$$\text{Minimize} \quad \omega \tag{2}$$

Subject to :

$$\sum_{j\in V_{\mathcal{H}}} x^T_{0jk} \leq 1 \qquad k\in\mathcal{K} \tag{3}$$

$$\sum_{j\in V_{\mathcal{H}}} x^T_{0jk} = \sum_{j\in V_{\mathcal{H}}} x^T_{j,n+1,k} \qquad k\in\mathcal{K} \tag{4}$$

$$\sum_{\substack{j\in V_{\mathcal{L}},\\ j\neq i}} x^T_{jik} = \sum_{\substack{j\in V_{\mathcal{R}},\\ j\neq i}} x^T_{ijk} \qquad k\in\mathcal{K}, i\in V_{\mathcal{H}} \tag{5}$$

$$\sum_{i\in V_{\mathcal{L}}}\sum_{\substack{j\in V_{\mathcal{N}},\\ j\neq i}}\sum_{\substack{h\in V,\\ h\neq j}} x^D_{\substack{ijh\\pk}} \leq 1 \qquad p\in\mathcal{P}, k\in\mathcal{K} \tag{6}$$

$$\sum_{\substack{j\in V_{\mathcal{N}},\\ j\neq i}}\sum_{\substack{h\in V,\\ h\neq j}} x^D_{\substack{ijh\\pk}} \leq \sum_{\substack{j\in V_{\mathcal{R}},\\ j\neq i}} x^T_{ijk} \qquad i\in V_{\mathcal{L}}, p\in\mathcal{P}, k\in\mathcal{K} \tag{7}$$

$$\sum_{i\in V_{\mathcal{L}}}\sum_{\substack{j\in V_{\mathcal{N}},\\ j\neq h,\\ j\neq i}} x^{D}_{\substack{ijh\\pk}} \le \sum_{\substack{i\in V_{\mathcal{L}},\\ i\neq h}} x^{T}_{ihk} \qquad h\in V_{\mathcal{R}}, p\in\mathcal{P}, k\in\mathcal{K} \tag{8}$$

$$\sum_{i\in V_{\mathcal{L}}}\sum_{\substack{j\in V_{\mathcal{N}},\\ j\neq i}}\sum_{\substack{h\in V,\\ h\neq j}} x^{D}_{\substack{ijh,\\p+1,k}} \le \sum_{i\in V_{\mathcal{L}}}\sum_{\substack{j\in V_{\mathcal{N}},\\ j\neq i}}\sum_{\substack{h\in V,\\ h\neq j}} x^{D}_{\substack{ijh\\pk}} \qquad p\in\mathcal{P}\setminus\{P\}, k\in\mathcal{K} \tag{9}$$

$$\sum_{k\in\mathcal{K}}\sum_{\substack{j\in V_{\mathcal{L}},\\ j\neq i}} x^{T}_{jik} \le 1 \qquad i\in V_{\mathcal{S}} \tag{10}$$

$$\sum_{p\in\mathcal{P}}\sum_{r\in\mathcal{R}}\sum_{j\in V_{\mathcal{N}}} x^{R}_{\substack{ijr\\pk}} \le n\cdot \sum_{\substack{h\in V_{\mathcal{L}},\\ h\neq i}} x^{T}_{hik} \qquad i\in V_{\mathcal{S}}, k\in\mathcal{K} \tag{11}$$

$$\sum_{\substack{h\in V_{\mathcal{L}},\\ h\neq i}} x^{T}_{hik} \le \sum_{p\in\mathcal{P}}\sum_{r\in\mathcal{R}}\sum_{j\in V_{\mathcal{N}}} x^{R}_{\substack{ijr\\pk}} \qquad i\in V_{\mathcal{S}}, k\in\mathcal{K} \tag{12}$$

Constraints (13) form the core of the delivery operations and guarantee the multimodality assignation, ensuring that each customer is served by exactly one transportation mode, i.e., either a truck, drone, or robot.

$$\sum_{k\in\mathcal{K}}\sum_{\substack{i\in V_{\mathcal{L}},\\ i\neq j}} x^{T}_{ijk} + \sum_{k\in\mathcal{K}}\sum_{p\in\mathcal{P}}\sum_{\substack{i\in V_{\mathcal{L}},\\ i\neq j}}\sum_{\substack{h\in V,\\ h\neq j}} x^{D}_{\substack{ijh\\pk}} + \sum_{k\in\mathcal{K}}\sum_{p\in\mathcal{P}}\sum_{r\in\mathcal{R}}\sum_{i\in V_{\mathcal{S}}} x^{R}_{\substack{ijr\\pk}} = 1 \qquad j\in V_{\mathcal{N}} \tag{13}$$

Constraints (14) establish that the total weight of parcels q_j that leave the depot on-board a truck k should respect the maximum truck capacity Q.

$$\sum_{i\in V_{\mathcal{L}}}\sum_{\substack{j\in V_{\mathcal{N}},\\ j\neq i}} x^{T}_{ijk}q_j + \sum_{p\in\mathcal{P}}\sum_{i\in V_{\mathcal{L}}}\sum_{\substack{j\in V_{\mathcal{N}},\\ j\neq i}}\sum_{\substack{h\in V,\\ h\neq j}} x^{D}_{\substack{ijh\\pk}}q_j + \sum_{p\in\mathcal{P}}\sum_{r\in\mathcal{R}}\sum_{i\in V_{\mathcal{S}}}\sum_{j\in V_{\mathcal{N}}} x^{R}_{\substack{ijr\\pk}}q_j \le Q \qquad k\in\mathcal{K} \tag{14}$$

Constraints (15)–(17) establish that every node has a specific position in the truck routes. Consequently, if a truck traverses arc (i,j), node j should be placed immediately after node i. In particular, Constraints (17) use the position of nodes in the trucks' routes to establish the order in which the pth drone sorties are executed. Thereby, if node j precedes node i in the truck route, the order of the $(p+1)$th sortie starting from j should be greater than the order of the pth sortie finishing at i.

$$z_{jk} - z_{ik} \ge 1-(n+2)(1-x^{T}_{ijk}) \qquad i\in V_{\mathcal{L}}, j\in V_{\mathcal{R}}, k\in\mathcal{K} : i\neq j \tag{15}$$

$$z_{jk} - z_{ik} \le 1+(n+2)(1-x^{T}_{ijk}) \qquad i\in V_{\mathcal{L}}, j\in V_{\mathcal{R}}, k\in\mathcal{K} : i\neq j \tag{16}$$

$$z_{jk} - z_{ik} \ge -M\Big(2-\sum_{h\in V_{\mathcal{L}}}\sum_{\substack{s\in V_{\mathcal{N}},\\ s\neq h,\\ s\neq i}} x^{D}_{\substack{hsi\\pk}} - \sum_{\substack{s\in V_{\mathcal{N}},\\ s\neq i,\\ s\neq j}}\sum_{\substack{h\in V,\\ h\neq s}} x^{D}_{\substack{jsh,\\p+1,k}}\Big) \qquad i\in V_{\mathcal{L}}, j\in V_{\mathcal{R}}, p\in\mathcal{P}\setminus\{P\}, k\in\mathcal{K} : i\neq j \tag{17}$$

Constraints (18)–(27) define the synchronization constraints for the different transportation modes. Constraints (18) define the arrival time of truck k at node j when traversing arc (i,j). Constraints (19) state that the starting time of the truck service is greater than its arrival time at i. Constraints (20) indicate that the truck departure time is greater than the starting time of the truck service at i. Constraints (21) state that if a drone sortie finishes at h, the truck departure time from h should be greater than the drone arrival time at h. Constraints (22) establish that the drone departure time from i should be greater than the truck arrival time at i plus the parcel-loading time. Constraints (23) establish that if a drone sortie starts from i, the truck departure time should be greater than the drone departure time from i. Constraints (24) guarantee that, if the drone traverses arc (i,j) to serve customer j, the arrival time at j is greater than the sum of the drone departure time from i and the travel time from i to j. Constraints (25) ensure that the starting departure time of the $(p+1)$th sortie is greater than the arrival time of the previous pth sortie. Constraints (26)–(27) define the release times for robot delivery operations.

$$a^{T}_{jk} \ge d^{T}_{ik} + t^{T}_{ij} - M(1-x^{T}_{ijk}) \qquad i\in V_{\mathcal{L}}, j\in V_{\mathcal{R}}, k\in\mathcal{K} : i\neq j \tag{18}$$

$$s^{T}_{ik} \ge a^{T}_{ik} - M\Big(1-\sum_{\substack{j\in V_{\mathcal{L}},\\ j\neq i}} x^{T}_{jik}\Big) \qquad i\in V_{\mathcal{R}}, k\in\mathcal{K} \tag{19}$$

$$d^{T}_{ik} \ge s^{T}_{ik} + \bar{l}^{T} - M\Big(1-\sum_{\substack{j\in V_{\mathcal{L}},\\ j\neq i}} x^{T}_{jik}\Big) \qquad i\in V_{\mathcal{R}}, k\in\mathcal{K} \tag{20}$$

$$d^{T}_{hk} \ge a^{D}_{jpk} + \bar{l}^{D} + t^{D}_{jh} - M\Big(1-\sum_{\substack{i\in V_{\mathcal{L}},\\ i\neq j}} x^{D}_{\substack{ijh\\pk}}\Big) \qquad j\in V_{\mathcal{N}}, h\in V, p\in\mathcal{P}, k\in\mathcal{K} : j\neq h \tag{21}$$

$$d^{D}_{ipk} \ge a^{T}_{ik} + l^{D} - M\Big(1-\sum_{\substack{j\in V_{\mathcal{N}},\\ j\neq i}}\sum_{\substack{h\in V,\\ h\neq j}} x^{D}_{\substack{ijh\\pk}}\Big) \qquad i\in V_{\mathcal{L}}, p\in\mathcal{P}, k\in\mathcal{K} \tag{22}$$

$$d^T_{ik} \geq d^D_{ipk} - M(1 - \sum_{\substack{j\in V_{\mathcal{N}},\\ j\neq i}} \sum_{\substack{h\in V,\\ h\neq j}} x^D_{\substack{ijh\\pk}}) \qquad i \in V_{\mathcal{L}}, p \in \mathcal{P}, k \in \mathcal{K} \tag{23}$$

$$a^D_{jpk} \geq d^D_{ipk} + t^D_{ij} - M(1 - \sum_{\substack{h\in V,\\ h\neq j}} x^D_{\substack{ijh\\pk}}) \qquad i \in V_{\mathcal{L}}, j \in V_{\mathcal{N}}, p \in \mathcal{P}, k \in \mathcal{K} : i \neq j \tag{24}$$

$$d^D_{i,p+1,k} \geq a^D_{jpk} + \bar{l}^D + t^D_{jh} + l^D - M(2 - \sum_{\substack{s\in V_{\mathcal{L}},\\ s\neq j}} x^D_{\substack{sjh\\pk}} - \sum_{\substack{s\in V_{\mathcal{N}},\\ s\neq h,\\ s\neq i}} \sum_{\substack{v\in V,\\ v\neq s}} x^D_{\substack{isv,\\ p+1,k}}) \qquad i, h \in V, j \in V_{\mathcal{N}}, p \in \mathcal{P} \setminus \{P\}, k \in \mathcal{K} : h \neq j, i \neq j \tag{25}$$

$$s^T_{ik} + \bar{l}^T \leq r^R_{\substack{ir\\0k}} + M(1 - \sum_{j\in V_{\mathcal{N}}} x^R_{\substack{ijr\\0k}}) \qquad i \in V_{\mathcal{S}}, r \in \mathcal{R}, k \in \mathcal{K} \tag{26}$$

$$r^R_{\substack{ir,\\ p-1,k}} + \sum_{j\in V_{\mathcal{N}}} (l^R + t^R_{ij} + \bar{l}^R + t^R_{ji}) x^R_{\substack{ijr,\\ p-1,k}} \leq r^R_{\substack{ir\\pk}} + M(1 - \sum_{j\in V_{\mathcal{N}}} x^R_{\substack{ijr\\pk}}) \qquad i \in V_{\mathcal{S}}, r \in \mathcal{R}, p \in \mathcal{P} \setminus \{0\}, k \in \mathcal{K} \tag{27}$$

Constraints (28)–(29) ensure that the drone-and-robot delivery operations are within the range that is allowed by the capacity of their batteries. Constraints (28) restrict the maximum flying time of drone a sortie from i to j to h, which is computed as the difference between the truck arrival time a^T_{hk} at h and the drone departure time d^D_{ipk} from node i. As a result, these constraints establish that the drone flying time from i to j to h, including hovering times at h, must be less than or equal to the time that the battery allows the drone to fly, that is, L_{ijh}. Constraints (29) restrict the maximum robot driving time for round-trips from i to j.

$$a^T_{hk} - d^D_{ipk} \leq L_{ijh} + M(1 - x^D_{\substack{ijh\\pk}}) \qquad i \in V_{\mathcal{L}}, j \in V_{\mathcal{N}}, h \in V, p \in \mathcal{P}, k \in \mathcal{K} : j \neq i, h \neq j \tag{28}$$

$$(l^R + t^R_{ij} + \bar{l}^R + t^R_{ji}) x^R_{\substack{ijr\\pk}} \leq T \qquad i \in V_{\mathcal{S}}, j \in V_{\mathcal{N}}, r \in \mathcal{R}, p \in \mathcal{P}, k \in \mathcal{K} \tag{29}$$

Time window constraints are incorporated by Constraints (30)–(35). Accordingly, a specific time interval in which each customer $i \in V_{\mathcal{N}}$ can be served is defined by setting a starting t^{start}_i and ending t^{stop}_i time. Constraints (30)–(31) indicate that if a truck visits customer i, the truck delivers the parcel within the customer's time-window interval. Constraints (32)–(33) ensure that drones arrive and serve customers within their corresponding time windows. Similarly, Constraints (34)–(35) ensure that the arrival and service times of robots satisfy the time-window constraints.

$$t^{start}_i \leq s^T_{ik} + M(1 - \sum_{\substack{j\in V_{\mathcal{L}},\\ j\neq i}} x^T_{jik}) \qquad i \in V_{\mathcal{N}}, k \in \mathcal{K} \tag{30}$$

$$t^{stop}_i + M(1 - \sum_{\substack{j\in V_{\mathcal{L}},\\ j\neq i}} x^T_{jik}) \geq s^T_{ik} + \bar{l}^T \qquad i \in V_{\mathcal{N}}, k \in \mathcal{K} \tag{31}$$

$$t^{start}_j \leq d^D_{ipk} + t^D_{ij} + M(1 - \sum_{\substack{h\in V,\\ h\neq j}} x^D_{\substack{ijh\\pk}}) \qquad i \in V_{\mathcal{L}}, j \in V_{\mathcal{N}}, p \in \mathcal{P}, k \in \mathcal{K} : i \neq j \tag{32}$$

$$t^{stop}_j + M(1 - \sum_{\substack{h\in V,\\ h\neq j}} x^D_{\substack{ijh\\pk}}) \geq d^D_{ipk} + t^D_{ij} + \bar{l}^D \qquad i \in V_{\mathcal{L}}, j \in V_{\mathcal{N}}, p \in \mathcal{P}, k \in \mathcal{K} : i \neq j \tag{33}$$

$$t^{start}_j \leq r^R_{\substack{ir\\pk}} + l^R + t^R_{ij} + M(1 - x^R_{\substack{ijr\\pk}}) \qquad i \in V_{\mathcal{S}}, j \in V_{\mathcal{N}}, r \in \mathcal{R}, p \in \mathcal{P}, k \in \mathcal{K} \tag{34}$$

$$t^{stop}_j + M(1 - x^R_{\substack{ijr\\pk}}) \geq r^R_{\substack{ir\\pk}} + (l^R + t^R_{ij} + \bar{l}^R) x^R_{\substack{ijr\\pk}} \qquad i \in V_{\mathcal{S}}, j \in V_{\mathcal{N}}, r \in \mathcal{R}, p \in \mathcal{P}, k \in \mathcal{K} \tag{35}$$

To reduce solutions that present the same routing, Constraints (36)–(38) break routes' symmetries.

$$\sum_{j\in V_{\mathcal{R}}} x^T_{0jk} \leq \sum_{j\in V_{\mathcal{R}}} x^T_{0j,k-1} \qquad k \in \mathcal{K} \setminus \{0\} \tag{36}$$

$$\sum_{j\in V_{\mathcal{N}}} x^R_{\substack{ijr\\pk}} \leq \sum_{j\in V_{\mathcal{N}}} x^R_{\substack{ijr\\p-1,k}} \qquad i \in V_{\mathcal{S}}, r \in \mathcal{R}, p \in \mathcal{P} \setminus \{0\}, k \in \mathcal{K} \tag{37}$$

$$\sum_{j\in V_{\mathcal{N}}} x^R_{\substack{ijr\\0k}} \leq \sum_{j\in V_{\mathcal{N}}} x^R_{\substack{ij,r-1\\0k}} \qquad i \in V_{\mathcal{S}}, r \in \mathcal{R} \setminus \{0\}, k \in \mathcal{K} \tag{38}$$

Pre-processing constraints are defined by Constraints (39)–(43). Constraints (39) forbid drone sorties that exceed the battery power. Constraints (40) prevent sorties that start from $i \in V_{\mathcal{R}}$ and finish at the starting depot $\{0\}$. Constraints (41)–(43) forbid sorties that involve nodes located in a no-fly zone. Note that additional pre-processing constraints based on the time windows can be developed. Hence, for customers $i, j \in V_{\mathcal{N}}$, where $t^{stop}_i < t^{start}_j$, customer j cannot be served first than customer i.

$$x^D_{\substack{ijh\\pk}} = 0 \qquad i \in V_{\mathcal{L}}, j \in V_{\mathcal{N}}, h \in V, p \in \mathcal{P}, k \in \mathcal{K} : j \neq i, h \neq j, b^{on}_{ij} + b^{off}_{jh} > B \tag{39}$$

$$x^{D}_{ij0pk} = 0 \qquad i \in V_{\mathcal{R}}, j \in V_{\mathcal{N}}, p \in \mathcal{P}, k \in \mathcal{K} \tag{40}$$

$$x^{D}_{ijhpk} = 0 \qquad i \in \mathcal{I}, j \in V_{\mathcal{N}}, h \in V, p \in \mathcal{P}, k \in \mathcal{K} \tag{41}$$

$$x^{D}_{ijhpk} = 0 \qquad i \in V_{\mathcal{L}}, j \in \mathcal{I}, h \in V, p \in \mathcal{P}, k \in \mathcal{K} \tag{42}$$

$$x^{D}_{ijhpk} = 0 \qquad i \in V_{\mathcal{L}}, j \in V_{\mathcal{N}}, h \in \mathcal{I}, p \in \mathcal{P}, k \in \mathcal{K} \tag{43}$$

The VRPD-RS-TW can also be used to minimize the makespan. For this, we define variable $\tau \in \mathbb{R}^{+}_{0}$, which represents the maximum of the earliest departure times for all trucks from $\{n+1\}$. Then, the objective function (1) is replaced by the makespan objective function (44), and Constraints (45) are incorporated into the MILP formulation.

$$\text{Minimize} \quad \tau \tag{44}$$

$$d^{T}_{n+1,k} \leq \tau \qquad k \in \mathcal{K} \tag{45}$$

Finally, valid inequalities for the MILP formulations of the makespan and operational costs VRPD-RS-TW are presented in Supplementary Material A.

4. Metaheuristic algorithm

As a result of the computational hardness intrinsically involved in the VRPD-RS-TW, we propose a Three-Phased Granular Multi-Start Iterated Local Search (3P-GMS-ILS) algorithm to solve the VRPD-RS-TW. A concise description of the 3P-GMS-ILS and its phases is presented in Section 4.1. A detailed description of how to compute the arrival, service, release, and departure times of trucks, drones, and robots is introduced in Section 4.2. Finally, the neighborhood structures and shaking procedures are described in Section 4.3.

4.1. Three-phased granular multi-start ILS algorithm

In this research, we develop a Three-Phased ILS framework that combines a multi-start mechanism with granular neighborhoods. The 3P-GMS-ILS consists of three phases. In Phase I (see Section 4.1.1), an initial feasible solution is built. After this, Phase II (see Section 4.1.3) and Phase III (see Section 4.1.2) are applied iteratively until the termination criterion is met. Fig. 2 presents a general scheme of the 3P-GMS-ILS, which is described in detail in Algorithm 1.

The 3P-GMS-ILS implements two successful algorithmic techniques to improve the overall performance of the metaheuristic framework: (*i*) a multi-start mechanism and (*ii*) granular neighborhoods. In the diversification phase, i.e., Phase II, the 3P-GMS-ILS implements a *multi-start mechanism* that utilizes a *re-starting list* to store the incumbent solutions identified during the search. Once a predefined diversification criterion is met, the search is re-initiated from these stored local optima. Inspired by the strategy proposed in Campuzano et al. (2021), this mechanism applies perturbation procedures over the solutions from the *re-starting list* to re-explore promising regions of the solution space. Unlike conventional multi-start approaches that often restart from random or initial low-quality solutions, this mechanism strategically leverages previously identified high-quality incumbents. This targeted re-exploration enables the algorithm to focus on previously identified promising areas rather than reverting to arbitrary or poor-quality solutions. Therefore, by activating the *multi-start mechanism* multiple times throughout the search process, 3P-GMS-ILS enhances its ability to identify improved incumbents while mitigating the effects of premature convergence. In addition to this procedure, the 3P-GMS-ILS utilizes *granular neighborhoods* in the intensification mechanism, i.e., Phase III, to drastically restrict and lead the exploration toward high-quality solutions (Toth and Vigo, 2003). Therefore, given a set of *granular arcs* $A_g \subseteq A$, defined as $A_g = \{(i,j) \in A : c^{t}_{ij} \leq \vartheta\}$, only truck arcs with transportation costs below the *granular threshold* ϑ are incorporated in A_g. The granular threshold is computed as shown in Eq. (46), where $A^{T}(S)$ represents the truck arcs of a given solution S, and β is the *sparsification parameter*. Here, the sparsification parameter β controls the level of selectivity when exploring candidate solutions or neighbors (Escobar et al., 2014). Thus, β limits the size of the neighborhood by filtering out less promising candidate solutions, whereas helping to reduce the computational effort required to explore the solution space of potentially good quality solutions (Shahmanzari and Aksen, 2021). It is worth noticing that granular neighborhoods can generate solutions with arcs that do not belong to A_g. This is due to neighborhood operators that may generate multiple new arcs. Consequently, a neighborhood operator is applied on a given solution S if at least one of the new arcs belongs to A_g. Moreover, in our implementation, drone and robot arcs are also incorporated in A_g. Hence, all drone arcs that belong to sorties (i,j,h) whose energy consumption does not exceed B are incorporated in A_g. Similarly, all robot arcs whose transportation times do not exceed T are added in A_g. Additionally, following the approach presented in Toth and Vigo (2003) and Schneider et al. (2017), all the edges incident to the depot are also incorporated in A_g.

$$\vartheta = \beta \cdot \frac{\sum_{(i,j)\in A^{T}(S)} c^{t}_{ij}}{(|V_{\mathcal{N}}| + |V_{\mathcal{S}}| + |\mathcal{K}|)} \tag{46}$$

Algorithm 1 describes the 3P-GMS-ILS. The input data of the 3P-GMS-ILS are the maximum number of multi-start procedures (R), the sparsification parameter (β), and the travel time matrices of trucks (t^{T}), drones (t^{D}), and robots (t^{R}). The algorithm's output is the best solution found in the search space S^{*} (incumbent). Further, $f(\cdot)$ provides the objective function value of the solutions, S_i stores the solution found during exploration, $RestartingList$ is a list that stores local optima, A_g is the set of granular arcs, counter

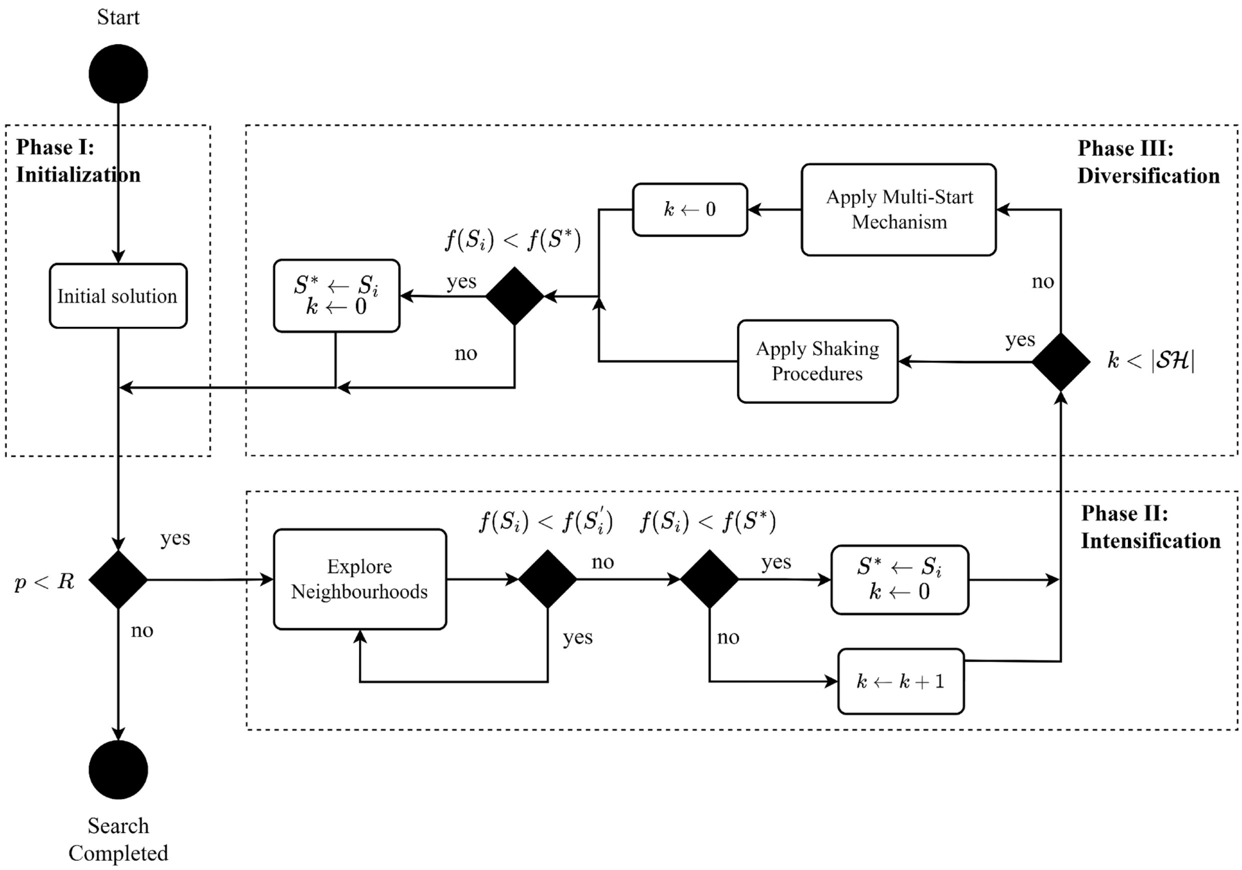


Fig. 2. General framework of the 3P-GMS-ILS metaheuristic.

i depicts the current iteration, counter p shows the number of shaking procedures that have been applied, and counter k identifies the shaking procedure that the algorithm should apply. Additionally, the 3P-GMS-ILS implements two functions, **AddEdges(·)** and **ResetEdges(·)**, designed to iteratively manipulate the total number of arcs in the set A_g. The **AddEdges(·)** function receives a feasible solution S as input parameter and incorporates all the arcs from S into A_g. The **ResetEdges(·)** function accepts a feasible solution S and the sparsification parameter β as input parameters, where β is used to compute the granular threshold ϑ. This function first clears A_g by setting $A_g = \emptyset$, then adds all the arcs from the set A below the *granular threshold* ϑ into A_g, and finally calls the **AddEdges(·)** function. Overall, the structure of the algorithm consists of two stages split into three phases. The first phase is the initialization, in which function **Phase_I(·)** builds an initial VRPD-RS-TW solution (line 1). Then, the **ResetEdges(·)** function guarantees that the set of granular arcs A_g is built, incorporating all the arcs of the initial solution S_i into A_g (line 2). In the second stage (lines 4–8), the diversification (line 5) and intensification (line 6) mechanisms are iteratively applied until the termination criterion is met, that is, p equals R (line 4). Note that the incumbent S^* is updated within Phases II and III.

Algorithm 1: Three-phased granular multi-start iterated local search.

```
   Data: (R, β, t^T, t^D, t^R)
1  S_i, S* ← Phase_I(t^T, t^D, t^R)
2  A_g ← ResetEdges(β, S_i)
3  i, p ← 0
4  while p < R do
5  │  S_i ← Phase_II(RestartingList, A_g, β, S*, S_i, t^T, t^D, t^R, k, p)
6  │  S_i ← Phase_III(A_g, β, S*, S_i, t^T, t^D, t^R, k)
7  │  S_{i+1} ← S_i
8  └  i ← i + 1
   Result: (S*)
```

4.1.1. Phase I: Initialization

In the first phase, an initial solution for the VRPS-RS-TW is built. This phase is described by Algorithm 2, where the input data are the truck, drone, and robot time matrices, i.e., t^T, t^D, and t^R, respectively. The algorithm first builds a truck-only solution by implementing the **SolveVRP(⋅)** function (line 1). This function determines the number of routes as: $routes = \lceil \frac{\sum_{i \in V_{\mathcal{N}}} q_i}{Q} \rceil$, where Q is the maximum capacity of the trucks and q_i is the parcel weight of customer $i \in V_{\mathcal{N}}$. The algorithm then iteratively searches for the farthest node in the direction $d = \frac{360°}{routes}$ until all the nodes have been assigned to a route. Here, trucks perform the routing over customers and robots stations, even though all the customers are initially served by trucks. After that, the **InsertRobots(⋅)** function is called to remove customer nodes from truck routes and assign them to robot stations following the cheapest insertion rule (line 2). Next, in each route, the **MakeFly(⋅)** function looks for three consecutive nodes visited by trucks (i, j, k) and evaluates a *sortie* where the drone now serves node j. The new sortie is accepted if the transportation costs are reduced, that is, if the following inequality is satisfied: $c^t_{i,k} + c^d_{i,j} + c^d_{j,k} < c^t_{i,j} + c^t_{j,k}$, where $c^t_{i,k} + c^d_{i,j} + c^d_{j,k}$ represents the costs of the new sortie and $c^t_{i,j} + c^t_{j,k}$ the costs of the previous delivery operations, i.e., visiting the three nodes by truck. The **MakeFly(⋅)** function is applied iteratively until no more three combined nodes can be turned into a sortie or no additional sortie meets the problem constraints. Then, the algorithm returns the initial solution S.

Algorithm 2: Phase I. Initialization.

Data: (t^T, t^D, t^R)
1 $S \leftarrow$ **SolveVRP**(t^T)
2 $S \leftarrow$ **InsertRobots**(t^R)
3 $S \leftarrow$ **MakeFly**(t^D)
Result: (S)

4.1.2. Phase II: Diversification

The second phase is implemented as a mechanism to escape from local optima and diversify the exploration. This phase is displayed by Algorithm 3 and its shaking procedures are described in Section 4.3. The input data are the restarting list in which local optima are stored $RestartingList()$, the granular arcs A_g, the sparsification parameter β, the incumbent S^*, the local optimum S_i, the travel time matrices (t^T, t^D, and t^R), the counter k corresponding to the shaking operations, and the counter p corresponding to the restarting operations. The output is a new modified solution S_i. Furthermore, $\mathcal{SH}_k(\cdot)$ represents the set of shaking structures, and $|\mathcal{SH}|$ is the total number of shaking procedures. Algorithm 3 starts by evaluating the value of k (line 1). If k is smaller than $|\mathcal{SH}|$, then the shaking

Algorithm 3: Phase II. Diversification.

Data: $(RestartingList(), A_g, \beta, S^*, S_i, t^T, t^D, t^R, k, p)$
1 **if** $k < |\mathcal{SH}|$ **then**
2   **while** $k < |\mathcal{SH}|$ **do**
3   **if** $\mathcal{SH}_k(S_i) = feasible$ **then**
4             $RestartingList() \leftarrow$ **Store**(S_i)
5             $S_i \leftarrow \mathcal{SH}_k(S_i)$
6   **break**
7   **else**
8             $k \leftarrow k + 1$
9 **else**
10     $S_i' \leftarrow S_i$
11     $S_i \leftarrow RestartingList()$
12     $k \leftarrow$ **Random()**
13     $S_i \leftarrow \mathcal{SH}_k(S_i)$
14     $RestartingList() \leftarrow$ **Delete**(S_i)
15     $RestartingList() \leftarrow$ **Store**(S_i')
16     $p \leftarrow p + 1$
17     $k \leftarrow 0$
18 $A_g \leftarrow$**ResetEdges**(β, S_i)
19 **if** $f(S_i) < f(S^*)$ **then**
20     $S^* \leftarrow S_i$
Result: (S_i)

procedures are applied (lines 2–8). Conversely, if k equals $|\mathcal{SH}|$ and the objective function has not been improved, then the algorithm changes strategy and activates the multi-start mechanism to re-explore promising regions of the solution space (lines 9–17). When implementing the shaking procedures, the algorithm begins its search by applying kth shaking procedure (line 2). If the kth shaking procedure $\mathcal{SH}_k(\cdot)$ generates a feasible solution, the current local optimum S_i is stored in the $RestartingList()$, the incumbent S_i is updated, and the algorithm goes out of the while-loop (lines 3–6). Otherwise, if kth shaking procedure $\mathcal{SH}_k(\cdot)$ produces an unfeasible solution, k is increased by one and the next shaking procedure is applied (lines 7–8). When applying the multi-start procedure, the algorithm first saves the current solution S_i in $S_i^{'}$ (line 10), and then a solution already stored in the restarting list is given to S_i (line 11). Afterward, a randomly chosen kth shaking procedure is applied to the new solution in S_i (lines 12–13), the solution S_i taken from the restarting list is deleted from *RestartinList()* (line 14), and the previously-found local optimum is stored in the restarting list (line 15). After this, the counter p is increased by one, and k is set to zero (lines 16–17). Next, after applying either the shaking procedures or the multi-start mechanism, the **ResetEdges(·)** function clears out set of granular arcs A_g and the arcs from solution S_i below the granular threshold are incorporated in A_g (line 18). Then, if the new solution in S_i improves the incumbent, i.e., $f(S_i) < f(S^*)$, the algorithm updates S^* (lines 19–20).

4.1.3. Phase III: Intensification

The third phase intensifies the search in the current solution's space by applying the set of neighborhood structures on a given solution. Phase III is displayed by Algorithm 4, and its neighborhoods are described in Section 4.3. The input data are the granular arcs A_g, the sparsification parameter β, the incumbent S^*, the current solution S_i, the travel time matrices (t^T, t^D, and t^R), and the counter k. In this phase, $S_i^{'}$ is utilized to save the values of S_i before the solution is adjusted during the exploration of the neighborhoods $\mathcal{N}_l(\cdot)$. Here, l identifies the lth neighborhood, and $|\mathcal{N}|$ represents the total number of neighborhoods. Algorithm 4

Algorithm 4: Phase III. Intensification.

Data: (A_g, β, S^*, S_i, t^T, t^D, t^R, k)
1 $f(S_i^{'}) \leftarrow \infty$
2 **while** $f(S_i) < f(S_i^{'})$ **do**
3 　$S_i^{'} \leftarrow S_i$
4 　$\mathcal{N}(\cdot) \leftarrow$ **Randomize()**
5 　**for** $l \in |\mathcal{N}|$ **do**
6 　　$S_i \leftarrow \mathcal{N}_l(A_g, S_i)$
7 　　**if** $f(S_i) \leq f(S_i^{'})$ **then**
8 　　　$A_g \leftarrow$**AddEdges**(S_i)
9 　　　**break**
10 **if** $f(S_i) < f(S^*)$ **then**
11 　$A_g \leftarrow$**ResetEdges**(β, S_i)
12 　$S^* \leftarrow S_i$
13 　$k \leftarrow 0$
14 **else**
15 　$k \leftarrow k + 1$
Result: (S_i)

starts by assigning the infinite value to the objective function of solution $S_i^{'}$ (line 1). Then, the set of neighborhoods is iteratively explored until a local optimum is found (lines 2–9). In every iteration, the solution S_i is saved in $S_i^{'}$ (line 3) and the order in which the neighborhoods are explored is randomized (line 4). Then, the exploration of the neighborhood structures $\mathcal{N}_l(\cdot)$ is executed following the new randomized order (lines 5–9). Here, the search is performed exhaustively within each neighborhood $\mathcal{N}_l(\cdot)$, i.e., exploring all the possible combinations, but only considering operator movements that involve at least one arc in A_g. This procedure is applied until either the solution $S_i^{'}$ is improved, i.e., $f(S_i) < f(S_i^{'})$, or all the neighborhoods $\mathcal{N}_l(\cdot)$ have been explored, i.e., $l > |\mathcal{N}|$. If the solution $S_i^{'}$ is improved, the **AddEdges(·)** function incorporates all the edges of the new solution S_i in A_g, and the algorithm goes out of the for-loop (lines 8–9) After that, if a new local optimum is found, the **ResetEdges(·)** function clears out the set of granular arcs A_g and adds all the arcs of the new solution S_i in A_g, the incumbent S^* is updated, and k is set to zero (lines 10–13). Otherwise, k increased by one unit (lines 14–15).

4.2. Synchronization of vehicles' deliveries

Many combinations of truck, drone, and robot delivery operations can be generated in a feasible solution. This section explains how the vehicles' synchronization times are computed in the 3P-GMS-ILS algorithm. The equations presented in this section are applied to every new solution built by the neighborhood structures and shaking procedures. To ease the presentation, we make use of the notation introduced in Section 3 and drop the sub-indexes corresponding to trucks and drones $k \in \mathcal{K}$ and robots $r \in \mathcal{R}$.

Trucks can park and wait until the time window becomes available. Hence, the truck arrival times a_j^T are computed by Eq. (47). If the truck arrives after the starting time of the time window, the service time s_i^T is computed by Eq. (48). Otherwise, s_i^T is computed by Eq. (49). Truck departure times d_i^T are computed by Eq. (50), which depends on the service time s_i^T and the drone departure time d_i^D.

$$a_j^T = d_i^T + t_{ij}^T \qquad i, j \in V \tag{47}$$

$$s_i^T = a_i^T \qquad i \in V_{\mathcal{N}} : a_i^T \geq t_i^{start} \tag{48}$$

$$s_i^T = t_i^{start} \qquad i \in V_{\mathcal{N}} : a_i^T < t_i^{start} \tag{49}$$

$$d_i^T = \max\{s_i^T + \bar{l}^T, \ d_i^D\} \qquad i \in V \tag{50}$$

Drone travel times depend on the type of sortie the drone performs, i.e., multi-leg or round-trip delivery operation. For drone multi-leg operations, Eq. (51) determine the drone arrival time a_j^D at the meeting point j. If the drone travels on-board the truck, a_j^D is computed by Eq. (52). Then, Eq. (53) compute the drone departure time d_i^D for cases where the drone leaves a node i on-board the truck and no round-trips are performed from i. To simplify the notation, we assume that r_i^D represents the release time corresponding to the last round trip from i and $round$ is the corresponding time to perform the last round trip from i to k to i, i.e., $round = l^D + t_{ik}^D + \bar{l}^D + t_{ki}^D$. Therefore, if a drone leaves node i on-board the truck and at least one round trip is performed from i, the drone departure time d_i^D is computed by Eq. (54). Furthermore, drone departure times d_i^D depend on the starting time of the time window t_j^{start}, the truck arrival time a_i^T, and the drone release time r_i^D. This way, if the earliest-possible drone arrival time, i.e., $max\{\cdot\} + t_{ij}^D$, is equal or larger than t_j^{start} and no round-trip are performed from i, the drone departure time d_i^D is computed by Eq. (55). If the earliest-possible drone arrival time is equal or larger than t_j^{start} and at least one round trip is performed from i, the drone departure time d_i^D is computed by Eq. (56). Conversely, if the earliest-possible drone arrival time at j is smaller than t_j^{start}, the drone departure time d_i^D is computed by Eq. (57). Note that in the right-hand side of Eqs. (55)–(57), the function $max\{\cdot\}$ represents the earliest-possible drone departure time from i and is computed as $max\{\cdot\} = \max\{a_i^D + l^D, \ a_i^T + l^D\}$, when no round trips are performed from i. Otherwise, $max\{\cdot\} = \max\{r_i^D + round + l^D, \ a_i^T + l^D\}$ for cases where at least one round-trip sortie is performed from i.

$$a_j^D = d_i^D + t_{ij}^D \qquad i \in V_{\mathcal{N}}, j \in V_{\mathcal{R}} \tag{51}$$

$$a_j^D = a_j^T \qquad j \in V_{\mathcal{R}} \tag{52}$$

$$d_i^D = \max\{a_i^D, \ s_i^T + \bar{l}^T\} \qquad i \in V_{\mathcal{L}} \tag{53}$$

$$d_i^D = \max\{r_i^D + round, \ s_i^T + \bar{l}^T\} \qquad i \in V_{\mathcal{L}} \tag{54}$$

$$d_i^D = \max\{a_i^D + l^D, \ a_i^T + l^D\} \qquad i \in V_{\mathcal{L}}, j \in V_{\mathcal{N}} : \max\{\cdot\} + t_{ij}^D \geq t_j^{start} \tag{55}$$

$$d_i^D = \max\{r_i^D + round + l^D, \ a_i^T + l^D\} \qquad i \in V_{\mathcal{L}}, j \in V_{\mathcal{N}} : \max\{\cdot\} + t_{ij}^D \geq t_j^{start} \tag{56}$$

$$d_i^D = t_j^{start} - t_{ij}^D \qquad i \in V_{\mathcal{L}}, j \in V_{\mathcal{N}} : \max\{\cdot\} + t_{ij}^D < t_j^{start} \tag{57}$$

Drone round-trip sorties corresponding to the first round-trip, i.e., $p = 0$, are set by Eq. (58) for cases where the earliest-possible arrival time is larger than or equal to t_j^{start}. Eq. (59) are applied for cases where the earliest-possible arrival time is less than t_j^{start}. Note that, in the right hand side of Eqs. (58) and (59), $max\{a_i^D, \ a_i^T\}$ represents the earliest-possible drone release time. Additionally, $round = l^D + t_{ih}^D + \bar{l}^D + t_{hi}^D$ represents the time it takes to perform the $(p-1) - th$ round trip. Therefore, the drone release time for the $p - th$ round trip is determined by Eq. (60) if the earliest-possible drone arrival time is larger than or equal to t_j^{start}. Conversely, Eq. (61) are applied if the earliest-possible drone arrival time is less than t_j^{start}.

$$r_{i0}^D = \max\{a_i^D, \ a_i^T\} \qquad i \in V_{\mathcal{L}}, j \in V_{\mathcal{N}} : \max\{a_i^D, \ a_i^T\} + l^D + t_{ij}^D \geq t_j^{start} \tag{58}$$

$$r_{i0}^D = t_j^{start} - (l^D + t_{ij}^D) \qquad i \in V_{\mathcal{L}}, j \in V_{\mathcal{N}} : \max\{a_i^D, \ a_i^T\} + l^D + t_{ij}^D < t_j^{start} \tag{59}$$

$$r_{ip}^D = r_{ip-1}^D + round \qquad p \in \mathcal{P} \setminus \{0\}, i \in V_{\mathcal{L}}, j \in V_{\mathcal{N}} : r_{ip-1}^D + round + l^D + t_{ij}^D \geq t_j^{start} \tag{60}$$

$$r_{ip}^D = t_j^{start} - (l^D + t_{ij}^D) \qquad p \in \mathcal{P} \setminus \{0\}, i \in V_{\mathcal{L}}, j \in V_{\mathcal{N}} : r_{ip-1}^D + round + l^D + t_{ij}^D < t_j^{start} \tag{61}$$

Robot round trips also depend on the customer time windows. Eq. (62) are used to compute the first robot release time r_{i0}^R, for cases where the earliest-possible robot arrival time is larger than or equal to t_j^{start}. Note that the first robot release time can only happen once the truck has delivered the parcel at the robot station, that is, after $a_i^T + s_i^T$. Eq. (63) are applied if the earliest-possible robot arrival time is less than t_j^{start}. Then, for the upcoming robot deliveries where $p \neq 0$, Eq. (64) are applied if the earliest-possible robot arrival time is larger than or equal to t_j^{start}. Here, $round = l^R + t_{ik}^R + \bar{l}^R + t_{ki}^R$ represents the time to perform the $(p-1) - th$ round trip. Eq. (65) are applied otherwise.

$$r_{i0}^R = a_i^T + s_i^T \qquad i \in V_{\mathcal{S}}, j \in V_{\mathcal{N}} : a_i^T + s_i^T + l^R + t_{ij}^R \geq t_j^{start} \tag{62}$$

$$r_{i0}^R = t_j^{start} - (l^R + t_{ij}^R) \qquad i \in V_{\mathcal{S}}, j \in V_{\mathcal{N}} : a_i^T + s_i^T + l^R + t_{ij}^R < t_j^{start} \tag{63}$$

$$r_{ip}^R = r_{i,p-1}^R + round \qquad p \in \mathcal{P} \setminus \{0\}, i \in V_{\mathcal{S}}, j \in V_{\mathcal{N}} : r_{i,p-1}^R + round + l^R + t_{ij}^R \geq t_j^{start} \tag{64}$$

$$r_{ip}^R = t_j^{start} - (l^R + t_{ij}^R) \qquad p \in \mathcal{P} \setminus \{0\}, i \in V_{\mathcal{S}}, j \in V_{\mathcal{N}} : r_{i,p-1}^R + round + l^R + t_{ij}^R < t_j^{start} \tag{65}$$

4.3. Neighborhood structures and shaking procedures

This section describes the neighborhood structures and shaking procedures incorporated into the 3P-GMS-ILS algorithm of Section 4.1. We adapt the neighborhoods *Push left* and *Push right* from Agatz et al. (2018); and neighbourhoods *Two optimality, Exchange 1.1, Exchange 2.2*, and *Reinsertion* from de Freitas and Penna (2020). Additionally, neighborhoods *Exchange Drone Round Trips with Robot Stations*, *Exchange Robot Stations*, and *Dissemble Sortie* are developed in this research. We consider the following neighborhood structures:

- *Push left*: This movement can only be applied if a sequence of nodes labeled as D-C-C is found in the drone route. Then, the movement changes the label of the middle *combined* node into a *truck* node, i.e., the *combined* node is deleted from the drone route and belongs only to the truck route.
- *Push right*: This structure can only be applied if a sequence of nodes labeled as C-C-D is found in the drone route. Then, the movement changes the label of the middle *combined* node into a *truck* node, i.e., the *combined* node is deleted from the drone route and belongs only to the truck route.
- *Two optimality*: In this operator, two nodes r_i and r_j from the truck route are selected and the arcs (r_i, r_{i+1}) and (r_{j-1}, r_j) are deleted. Then, the route between r_i and r_j is inverted and connected by adding the arcs (r_i, r_{j-1}) and (r_{i+1}, r_j). This way, the truck route $\{r_i, r_{i+1}, \dots, r_{j-1}, r_j\}$ becomes $\{r_i, r_{j-1}, \dots, r_{i+1}, r_j\}$.
- *Exchange 1.1*: This neighborhood structure adjusts the route by swapping the position of two nodes. This movement is applied to swap customers between the same or different routes and between the same or different transportation modes, i.e., trucks, drones, and robots.
- *Exchange 2.2*: In this movement, two pairs of consecutive nodes are selected and swapped in the truck route. For the same pair of nodes, seven different combinations can be generated. As such, all the combinations are valued, and the combination that reduces the costs the most is chosen.
- *Reinsertion*: This operator is applied to insert customers into truck routes, drone round-trips, and robot round-trips. Hence, one customer is selected from a route and inserted into the route of one of the other transportation modes.
- *Exchange drone round trips with robot stations*: This neighborhood exchanges all the customers served by drone round-trips, from a combined node, with all the customers served by robots from a robot station.
- *Exchange robot stations*: This movement selects two different robot stations from either the same or different routes, and then swaps the stations and the corresponding customers.
- *Dissemble sortie*: This structure selects a sortie and evaluates the insertion of the drone node as a combined node into another position of either the same or a different route. If the selected sortie has truck nodes between the combined nodes, e.g., C-T-T-C, the truck nodes are labeled as combined nodes in the same position, that is, C-C-C-C.

The shaking procedures receive the incumbent and generate a random feasible solution without evaluating the objective function. We consider the following shaking procedures:

- *Shaking 1*: This operator swaps two nodes, customers, or robot stations that the truck visits on the same or different routes.
- *Shaking 2*: This structure takes one node that is visited by the truck, either a combined or truck node, and inserts it into another route.
- *Shaking 3*: This movement dissembles a drone sortie and randomly inserts the drone-labeled node in either the same or a different route.
- *Shaking 4*: This procedure looks for a robot station in the trucks' routes and places it into another random position in one of the truck routes.
- *Shaking 5*: This structure selects one customer that is visited by a drone round-trip and inserts it into a random position in one of the truck routes.
- *Shaking 6*: This movement selects one customer visited by a robot round-trip and inserts it into a random position in one of the truck routes.

To guarantee feasibility, the operators $\mathcal{N}_l(\cdot)$ and $\mathcal{SH}_k(\cdot)$ exclude any nodes in the no-fly zones $\mathcal{I}$ from $V_{\mathcal{N}}$, $V_{\mathcal{L}}$, and $V_{\mathcal{R}}$, when exploring neighboring solutions that generate new drone delivery operations. Hence, time-dependent variables are computed as shown in Section 4.2, where no drone delivery operation may take off, serve, or land within a no-fly zone $\mathcal{I}$.

Furthermore, each time a new solution is generated within the neighboring structures $\mathcal{N}_l(\cdot)$ and shaking procedures $\mathcal{SH}_k(\cdot)$, we apply the full set of feasibility checks listed in Supplementary Material B. We do this because any local change in the route may shift subsequent time variables and violate constraints. Consequently, we reevaluate every constraint on the entire route, including both modified and unmodified portions, after each neighbor move, rather than partition checks by neighboring operators or segments.

5. Numerical experiments

In this section, we present the numerical experiments on the VRPD-RS-TW. Section 5.1 provides a detailed description of the experimental setup. Section 5.2 presents the results of four sets of experiments where we analyze the effectiveness of the VRPD-RS-TW using a case study of the city of Amsterdam, the Netherlands.

Table 3
Values of vehicles' parameters.

Parameter	Coefficient	Value	Unit	Reference
Drone speed	v^D	8.33	$[m/s]$	Campuzano et al. (2021)
Parcel weight	q	[0.1 − 8.0]	$[kg]$	Chen et al. (2021)
Time window range	t^{start}, t^{stop}	3	$[h]$	Chen et al. (2021)
Drone takeoff time	t^{tk}	60	$[s]$	Murray and Raj (2020)
Drone landing time	t^{l}	30	$[s]$	Murray and Raj (2020)
Truck service time	$\bar{t}^T$	30	$[s]$	Murray and Raj (2020)
Drone service time	$\bar{t}^D$	60	$[s]$	Murray and Raj (2020)
Robot service time	$\bar{t}^R$	40	$[s]$	Heimfarth et al. (2022)
Drone battery endurance	B	600	$[KJ]$	Campuzano et al. (2021)
Robot delivery endurance	T	3600	$[s]$	Starship Robot

5.1. Experimental design

This section describes the experimental settings, generation of realistic instances, and the settings concerning vehicles' travel times, operational costs, and drone energy consumption.

5.1.1. Computational settings and vehicle parameters

The computational experiments for the MILP formulations were conducted on a Microsoft Azure F32s v2 virtual machine, equipped with 32 virtual CPUs, 64 GB of RAM, and running Windows Server 2019. Additionally, the metaheuristic algorithms were executed on a Microsoft Azure E2ads v5 virtual machine, equipped with 2 virtual CPUs, 16 GB of RAM, and running Windows Server 2019. The mixed-integer linear programming model was coded in C+ + and solved using CPLEX 22.1.0. All the instances evaluated for the MILP model were run with a time limit of 7200 seconds.

The metaheuristic approach proposed in Section 4 needs two parameters: the maximum number of multi-start procedures (R) and the sparsification parameter (β). We evaluated 3P-GMS-ILS performance on 15 representative instances. For each parameter combination of $R \in \{5, 10, 15, 20, 25\}$ and $\beta \in \{1.0, 1.5, 2.0, 2.5, 3.0, 3.5\}$, we performed 10 independent runs per instance, normalizing objective values to account for varying instance sizes. To identify statistically significant differences among configurations, we applied the non-parametric Friedman test (Oda et al., 2015) at a significance level of $\alpha_{friedman} = 0.05$, and followed this with the post-hoc Nemenyi test (Liu and Chen, 2012) for pairwise comparisons. The best performance was obtained with $R = 25$ and $\beta = 3.0$, which yielded a mean normalized objective of 1.0594 (SD = 0.1199) and a 95% confidence interval of $[0.9930, 1.1258]$. The Friedman test returned an small p-value ($p \approx 1.0561374 \cdot 10^{-23}$), confirming highly significant differences among parameter configurations. These settings are adopted for all the subsequent experiments of this paper, where the best results across the 10 executions are reported. Additional details on the statistical analysis for the calibration of the metaheuristic can be found in Supplementary Material C.

The VRPD-RS-TW is a new extension of the VRPTW. For this reason, there are no available benchmark instances in the literature. Consequently, we created a new set of instances to model a real-case scenario based on the city of Amsterdam, the Netherlands, representing a densely urbanized city covering an area of 219 km^2. Additionally, we created synthetic instances with an area of 400 km^2, significantly larger than the Amsterdam case. While the Amsterdam instances capture the challenges of dense and compact city centers, the synthetic instances are specifically designed to reflect broader and more diverse urban settings, including low-density areas and hilly terrain. This helps us evaluate the optimization approach's adaptability in contexts with longer travel distances, fewer delivery points per area, and potential topographic variations. See Section 5.1.2 for details on the instances' generation. The drone and robot parameters are based on the Avidrone 210TL and the Starship Robot, respectively. Table 3 provides the considered parameter values for the experiments and their references.

5.1.2. Instance generation

The experiments are carried out on a generated set of instances for the city of Amsterdam, the Netherlands, and a synthetic set of instances representing a much larger area.

To create the Amsterdam instances, we used the software ArcGIS with map data provided by HERE Technologies. We randomly selected customer locations to create two sets of instances denoted as *small* and *large rings of Amsterdam*. The small-ring set consists of 3 different instances of 5, 10, and 15 customer locations in the district of Amsterdam Center. Similarly, the large-ring set consists of 3 different instances of 50, 75, and 100 customer locations in the districts of Amsterdam Center, North, East, West, and South. Additionally, 1, 2, 3, 10, 15, and 20 nodes were selected to belong to no-fly zones for the instances of 5, 10, 15, 50, 75, and 100 customers, respectively. The customer locations were randomly selected from the residential addresses in Amsterdam, respecting the relative distribution of these addresses over the different districts. Fig. 3 illustrates the districts considered to create the Amsterdam instance-set. The red marker indicates the depot's location, while the other markers represent the robot stations. We used the fastest paths in ArcGIS to calculate distances and travel times. Trucks' distances and travel times are based on urban cars, whereas robots' distances and travel times are based on walking speeds.

To assess the optimization approaches' adaptability, we created a synthetic set consisting of instances of 50 customers size, where the nodes $x - y$ coordinates are uniformly distributed in a square area of 20x20 [km]. These instances capture a vastly different, larger urban layout than Amsterdam's circular instances, incorporating long arcs through low-density areas and hilly terrain typical of less

Fig. 3. Central depot and robot stations for the generated instance sets of Amsterdam, the Netherlands. Source: OpenStreetMap.

densely populated last-mile scenarios. For this dataset alone, we assumed an average truck speed of 25 $[km/h]$ to reflect the maximum allowed speed within cities in the Netherlands. Then distances are calculated using Manhattan metrics.

For both sets of instances, the drone travel times of horizontal cruise are computed as $t^h_{ij} = \frac{h_{ij}}{v^D_{ij}}$ for each arc $(i,j) \in A$. Here, h_{ij} represents the great-circle distance between two locations i and j, and v^D_{ij} the drone speed. To compute h_{ij}, we apply the Haversine formula with a mean earth radius value of $r = 6371$ $[km]$ (Suryana et al., 2018). The new sets of instances are available at https://github.com/campuzanogf/VRPD-Robot-Stations-Time-Windows.

5.1.3. Operational costs computation

The operational costs presented in Section 3.2 consist of the routing costs and the drivers' wages that represent the total working time. The routing costs per kilometer of trucks, drones, and robots are given by c^T, c^D, and c^R, respectively.

The truck operational costs presented in the objective function ω consist of two components: the drivers' wage $f^T = 30.00 \cdot \frac{1}{3600}$ [€/s] and the truck routing costs c^T [€/km]. Note that f^T is multiplied by $d^T_{n+1,k}$, which is a variable measured in seconds and represents the time at which every truck $k \in \mathcal{K}$ finishes the delivery operations at the depot, i.e., node $\{n+1\}$. Moreover, drone and robot routing costs not only represent the incurred costs during transportation but also other costs intrinsically involved in delivery systems that incorporate autonomous vehicles. These costs include maintenance, remote piloting, insurance, battery degradation, and depreciation. Therefore, in the first three sets of experiments (see Sections 5.2.1, 5.2.3, and 5.2.4), we consider truck routing costs of $c^T = 0.25$ [€/km], and drone and robot routing costs of $c^D = c^R = 1.00$ [€/km]. Here, due to the similarity of the additional costs of drones and robots, we assume the same routing costs for drones and robots. In future scenarios where drone and robot delivery systems achieve widespread adoption, significant cost reductions are anticipated as a result of economies of scale, technological advancements, and increased market competition. factors are expected to decrease the operational costs of these emerging technologies, potentially bringing their routing costs closer to those of traditional trucks. To explore the implications of such potential scenarios, Section 5.2.5 presents a sensitivity analysis in which we assume equal routing costs for all delivery modes, i.e., $c^T = c^D = c^R = 0.25$ [€/km]. This approach not only enables us to assess how the delivery network's performance and structure might evolve under more cost-efficient conditions but also serves to validate the results obtained in Section 5.2.4 across different cost configurations. Furthermore, the different cost settings assessed for trucks, drones, and robots in Sections 5.2.4 and 5.2.5 align with the cost values presented in Ostermeier et al. (2021), Heimfarth et al. (2022), Nguyen et al. (2022), and Ostermeier et al. (2023).

5.1.4. Drone battery consumption model

When the drone traverses arc $(i,j) \in A$ with a parcel q_j on board to serve customer $j \in V_{\mathcal{N}}$, the transportation time is computed as $t^D_{ij} = t^{tk} + t^h_{ij} + t^l$ (see Section 5.1.2 for details on t^{tk}, t^h, and t^l). We follow the approach presented in Murray and Raj (2020), Raj and Murray (2020); and Campuzano et al. (2023) to compute the energy consumption model as $b_{ij} = t^{tk} \cdot p^{tl}_i(q_j, v_{ve}) + t^h_{ij} \cdot p^c_{ij}(q_j, v^D_{ij}) + t^l \cdot p^{tl}_j(q_j, v_{ve})$ (Liu et al., 2017). Here, p^{tl} is the induced power of the vertical flow for either taking off or landing, p^c represents the power consumption during the horizontal cruise, and p^h determines the energy consumed during hover. We implement b_{ij} to compute b^{on}_{ij} and b^{off}_{ij} for the case where the drone traverses arc $(i,j) \in A$ with a parcel on- and off-board, respectively. Notice that the parcel weight is set to $q_j = 0$ to compute b^{off}_{ij}.

We restrict the drone flying range by incorporating the hovering time into the maximum flying time L_{ijh}, for a delivery operation from i to j to h. The flying time of a sortie is computed as $t^{sortie}_{ijh} = t^D_{ij} + \bar{l}^D + t^D_{jh}$ and the energy consumption as $b^{sortie}_{ijh} = b^{on}_{ij} + b^{off}_{jh}$. It is worth noticing that t^{sortie}_{ijh} does not consider the hovering time at h. Consequently, the maximum time that a drone can hover is the remaining energy after finishing the delivery, i.e., $B - b^{sortie}_{ijh}$, divided by the consumption rate during hover without a parcel on board, i.e., $p^h_{(0)}$. Then, L_{ijh} is computed by Eq. (66).

$$L_{ijh} = t^{sortie}_{ijh} + \frac{B - b^{sortie}_{ijh}}{p^h_{(0)}} \qquad i,j,h \in V, b^{on}_{ij} + b^{off}_{jh} < B, i \neq j, j \neq h \tag{66}$$

5.2. Numerical results

This section provides the numerical results of comparing the MILP formulation, the heuristic algorithm, and the managerial analysis. The experiments of this paper are divided into four sets. The first set of experiments (Section 5.2.1) focuses on studying different objective functions to minimize operational costs. The second set of experiments (Section 5.2.2) evaluates the contribution of each component to the overall effectiveness of 3P-GMS-ILS. The third set of experiments (Section 5.2.3) compares different metaheuristic approaches and provides insights into the relationship between the operational costs and makespan objective functions. The fourth set of experiments (Section 5.2.4) analyzes the versatility of the VRPD-RS-TW by comparing this delivery system with variants that include fewer transportation modes. The fifth set of experiments (Section 5.2.5) studies the sensitivity of the routing costs of drones and robots for the VRPD-RS-TW and the related delivery systems. Finally, the sixth set of experiments (Section 5.2.6) provides a sensitivity analysis on the parameter configuration of the 3P-GMS-ILS metaheuristic.

5.2.1. Analysis of different operational costs objectives

The VRPD-RS-TW aims to minimize the operational costs of trucks, drones, and robots. In our MILP formulation, we consider drivers' wages to incorporate driving costs into the objective function. However, in this set of experiments, we study the operational costs objective function when including and excluding the drivers' wages to validate the 3P-GMS-ILS algorithm compared with the MILP formulation. Consequently, we solve instances of 5, 10, and 15 customers for the small ring of Amsterdam. Due to the size of the instances and the parcel weights, we set aside the capacity constraints, i.e., Constraints (14), and consider only one truck at the depot and one autonomous robot available at the station, i.e., $|\mathcal{K}| = 1$ and $|\mathcal{R}| = 1$. The results are listed in Tables 4 and 5, where column n provides the number of customers per instance, Z^W_{MILP} the best solution found for the MILP formulation when incorporating drivers' wages in the objective function, Z^R_{MILP} the best solution found for the MILP formulation when excluding drivers' wages from the objective function, Gap (%) the corresponding MILP optimality gap, Z_{3P} the best solution found for the 3P-GMS-ILS algorithm, Z_{Init} the objective function value of the initial solution of the 3P-GMS-ILS before phases II and III, ΔZ the percentage difference between Z_{MILP} and Z_{3P} computed as $\Delta Z = 100 \cdot \frac{Z_{MILP} - Z_{3P}}{Z_{MILP}}$, where Z_{MILP} corresponds to Z^W_{MILP} and Z^R_{MILP} in Tables 4 and 5, respectively, and ΔZ_{Init} the percentage difference between Z_{Init} and Z_{3P} computed as $\Delta Z_{Init} = 100 \cdot \frac{Z_{Init} - Z_{3P}}{Z_{Init}}$. The results presented in Tables 4

Table 4
Comparison of the MILP_{Val} and 3P-GMS-ILS when including the drivers' wages in the objective function.

Instance		MILP_{Val}				3P-GMS-ILS				
#	n	Z^W_{MILP}	Z_R	Gap [%]	Time [s]	Z_{3P}	Z_{Init}	ΔZ [%]	ΔZ_{Init} [%]	Time [s]
1		**132.23**	5.04	0.00	29.47	**132.23**	140.24	0.00	5.71	0.32
2	5	**57.35**	13.33	0.00	7.97	**57.35**	132.55	0.00	56.73	0.25
3		**151.39**	3.13	0.00	34.17	**151.39**	153.13	0.00	1.14	0.31
1		64.32	18.45	58.44	7204.58	**61.41**	162.29	4.52	62.16	0.95
2	10	134.01	9.24	84.60	7204.19	**133.89**	158.29	0.09	15.41	1.03
3		96.90	27.06	83.34	7225.36	**96.30**	159.81	0.62	39.74	0.96
1		122.47	15.93	94.66	7224.05	**107.13**	177.64	12.53	39.70	4.22
2	15	169.44	9.58	87.46	7205.47	**168.53**	179.59	0.53	6.15	4.14
3		132.75	11.22	93.96	7216.13	**89.38**	155.33	32.67	42.46	3.44
Avg		117.87	12.55	55.83	4816.82	**110.85**	157.65	5.66	29.91	1.73

and 5 correspond to the MILP formulations integrating the valid inequalities (MILP_{Val}) introduced in Supplementary Material A, which present slightly better results than the MILP in terms of objective function values, optimality gaps, and computational times. Furthermore, in Table 4, column Z_R shows the routing costs of Z^W_{MILP}, which are obtained by excluding the drivers' wages from the final result. Similarly, in Table 5, column Z_W reports the values of incorporating the drivers' wages into the results of Z^R_{MILP}. In the tables, costs are displayed in euros (€), and computational times are shown in seconds for the MILP_{Val} formulation and 3P-GMS-ILS metaheuristic under the column Time (s). Every instance was run 10 times for the 3P-GMS-ILS, and the best solutions found are reported. Best values are bold-faced.

Results in Tables 4 and 5 demonstrate that the 3P-GMS-ILS outperforms the results of the MILP_{Val} formulation for the small set of instances of Amsterdam. Columns ΔZ show that the 3P-GMS-ILS finds solutions that, on average, are 5.66 % and 0.00 % better than the MILP_{Val} formulation, for the case when including and excluding the drivers' wages, respectively. This validates the performance of 3P-GMS-ILS for small instances, as it consistently finds the same optimal solutions as MILP_{Val} for both objectives and achieves better objective values when MILP_{Val} does not prove optimality. Furthermore, columns ΔZ_{Init} show that the 3P-GMS-ILS effectively explores the solution space with final solutions that, on average, are 29.91 % and 40.46 % better than the initial solutions when including and excluding the drivers' wages. In particular, while we might expect higher average values for ΔZ_{Init} across both objectives, the results indicate otherwise. In Table 5, where both methods reach optimality, the low ΔZ_{init} values confirm that the initial solutions are already near-optimal solutions. Similarly, in Table 4, 3P-GMS-ILS only needs to improve its initial solutions by an average of 29.91 % to outperform the MILP_{Val} incumbents. This demonstrates that the modest average improvement achieved by 3P-GMS-ILS is not the result of limited exploration but rather reflects the high quality of its initial solutions.

When comparing the results of the MILP_{Val} formulation in Tables 4 and 5, we see that the VRPD-RS-TW becomes significantly more complex when including the drivers' wages in the objective function. As such, the MILP_{Val} formulation shows gaps of 55.83 % and 0.00 % for the cases when including and excluding the drivers' wages, respectively. Including drivers' wages (Z^W_{MILP}) displays a term that weights each vehicle's final time at the depot $d^T_{n+1,k}$ by the driving cost per hour f^T. Unlike the tight linear relaxation of summing edge costs (Z^R_{MILP}), the additional term $d^T_{n+1,k}$ in Z^W_{MILP} is computed from transportation times, service times, and waiting times multiplied by fractional values of x^T_{ijk}, which weakens the linear relaxation. This leads to a larger branch-and-bound tree search and a higher reported optimality gap. Nevertheless, when including drivers' wages, the relative gap between MILP_{Val} and the metaheuristic (ΔZ) is only 5.66 %, compared to a 55.83 % optimality gap. This discrepancy shows that the large MILP_{Val} gaps are due to the weak linear relaxation and not poor-quality incumbent solutions.

Additionally, when comparing the routing costs of Tables 4 and 5, i.e., Z_R and Z^R_{MILP}, respectively, Z^R_{MILP} displays better routing costs than Z_R. However, when comparing the operational costs of Tables 4 and 5, i.e., Z^W_{MILP} and Z_W, respectively, Z^W_{MILP} reports better operational costs than Z_W, even though Z^W_{MILP} shows average gaps of 55.83 % and Z_W shows average gaps of 00.0 %. Therefore, results show that, for the small set of instances of Amsterdam, the no-optimal solutions of minimizing operational costs (column Z^W_{MILP} in Table 4) outperform the operational costs values corresponding to the optimal solutions of minimizing routing costs (column Z_W in Table 5).

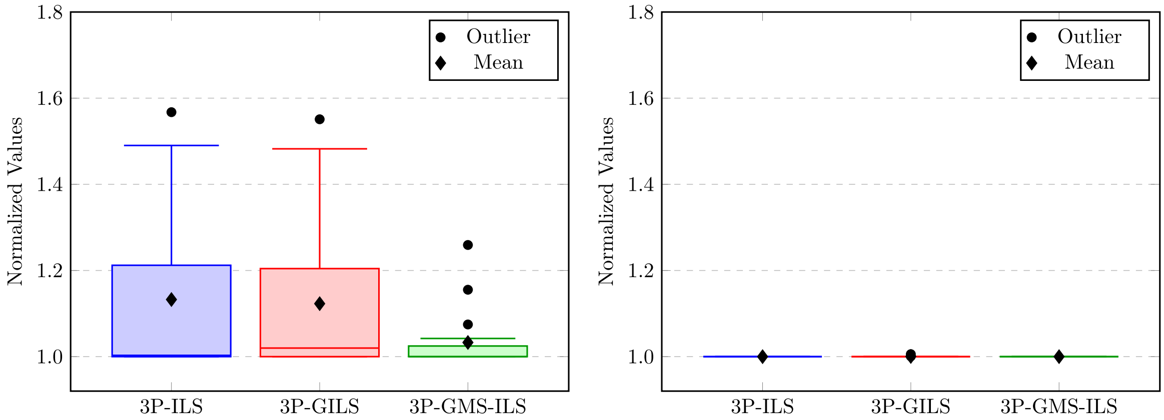
5.2.2. *Evaluation of components for the 3P-GMS-ILS metaheuristic*

These experiments evaluate the effectiveness of the components within the 3P-GMS-ILS metaheuristic. Specifically, we compare the full version to two simplified variants: (*i*) a version excluding the *multi-start* procedure referred to as 3P-GILS, and (*ii*) a version excluding both the *multi-start* procedure and the use of *granular neighborhoods* referred to as 3P-ILS. The experimental setup considers three trucks stationed at the depot and three autonomous robots available at each station, i.e., $|\mathcal{K}| = 3$ and $|\mathcal{R}| = 3$. Table 6 presents the objective function values and computational times in seconds for each metaheuristic when applied to the Amsterdam instance set. The best results for operational costs and makespan, in terms of both objective values and computational times, are highlighted in bold.

Supplementary Material E provides the results of the non-parametric Friedman test, used to evaluate performance differences among the 3P-ILS, 3P-GILS, and 3P-GMS-ILS metaheuristics. For the operational cost objective, the Friedman test yields p-values of

Table 5
Comparison of the $MILP_{Val}$ and 3P-GMS-ILS when excluding the drivers' wages from the objective function.

Instance		$MILP_{Val}$				3P-GMS-ILS				
#	n	Z^R_{MILP}	Z_W	Gap [%]	Time [s]	Z_{3P}	Z_{Init}	ΔZ [%]	ΔZ_{Init} [%]	Time [s]
1		**4.30**	142.09	0.00	0.48	**4.30**	6.63	0.00	35.16	0.23
2	5	**6.00**	131.58	0.00	1.19	**6.00**	6.96	0.00	13.90	0.22
3		**3.13**	151.39	0.00	1.08	**3.13**	4.38	0.00	28.69	0.15
1		**5.85**	171.79	0.00	847.72	**5.85**	9.08	0.00	35.61	0.88
2	10	**6.64**	157.27	0.00	484.34	**6.64**	11.0	0.00	39.61	0.76
3		**3.99**	159.15	0.00	43.52	**3.99**	6.94	0.00	42.52	0.70
1		**4.73**	237.34	0.00	1262.66	**4.73**	14.15	0.00	66.60	2.00
2	15	**8.50**	224.82	0.00	334.75	**8.50**	14.21	0.00	40.20	1.43
3		**4.43**	242.65	0.00	309.19	**4.43**	11.60	0.00	61.85	2.32
Avg		**5.28**	179.79	0.00	364.99	**5.28**	9.44	0.00	40.46	0.97



(a) Performance for the operational-costs objective (b) Performance for the makespan objective

Fig. 4. Distribution of normalized objective values for 3P-ILS, 3P-GILS, and 3P-GMS-ILS.

0.0004, i.e., below 0.05, indicating statistically significant differences among the metaheuristics. In contrast, the test shows p-values of 0.1353 for the makespan objective, i.e., exceeded 0.05, suggesting no significant differences in that case. The results visualized in Fig. 4 confirm that 3P-GMS-ILS consistently achieves the best performance for operational costs, with the lowest normalized values across the mean, standard deviation, standard error of the mean, and the narrowest confidence intervals, with values of 1.0329, 0.0688, 0.0162, and $[0.9987, 1.0671]$, respectively. Conversely, the lack of significant differences for the makespan objective implies that all three metaheuristics perform comparably well in minimizing this objective.

The results from Table 6 show that the best metaheuristic approach for the small and large set of instances of Amsterdam is the 3P-GMS-ILS. The 3P-GMS-ILS outperforms 3P-ILS and 3P-GILS with average objective values of 228.69 for the operational costs objective, whereas the three metaheuristics present the same results for the makespan objective. This indicates that operational costs are more complex to solve than the makespan objective. Furthermore, the results demonstrate that incorporating *granular neighborhoods* effectively reduces computational time. Specifically, 3P-GILS achieves lower computational times than 3P-ILS for both objective functions. This confirms the efficiency of granular neighborhoods, which eliminate large numbers of poor-quality arcs, thereby directing the search process and focusing exploration on more promising regions of the solution space. On the other hand, the results indicate that the *multi-start* mechanism is effective in escaping local optima and guiding the search toward high-quality regions of the solution space. As expected, 3P-GMS-ILS incurred higher computational times compared to 3P-ILS and 3P-GILS, due to the repeated re-initialization of the search process. However, this additional computational effort translates into improved performance for the operational cost objective. Therefore, these results highlight the effectiveness of combining *granular neighborhoods* and the *multi-start* mechanism in 3P-GMS-ILS for improving performance on the operational cost objective, while offering no significant advantage for the makespan objective.

5.2.3. Comparison of different metaheuristic approaches

These experiments compare the performance of 3P-GMS-ILS with the GVNS presented in Morim et al. (2024) and a Simulated Annealing (SA) metaheuristic for the Amsterdam set of instances. Table 7 contains the same columns and setup as Table 6. The SA pseudo-code and its parameters are shown in Supplementary Material D. Fig. 5 shows a box plot of the non-parametric analysis, illustrating the normalized performance metrics for both the makespan and operational costs objective functions. Additional

Table 6
Comparison of the 3P-GMS-ILS metaheuristic components for the operational costs and makespan objective functions.

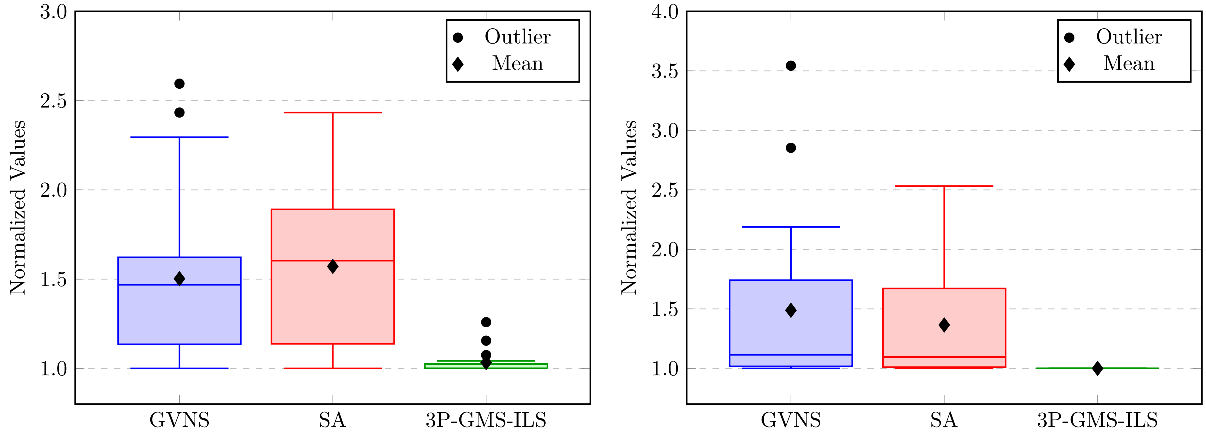
Instance		Operational Costs						Makespan					
		3P-ILS		3P-GILS		3P-GMS-ILS		3P-ILS		3P-GILS		3P-GMS-ILS	
#	n	ω	Time [s]	ω	Time [s]	ω	Time [s]	τ	Time [s]	τ	Time [s]	τ	Time [s]
1		**132.23**	**0.25**	**132.23**	0.26	**132.23**	0.32	**4.24**	**0.18**	**4.24**	**0.18**	**4.24**	0.25
2	5	**57.35**	0.12	**57.35**	**0.11**	**57.35**	0.25	**1.47**	0.21	**1.47**	**0.14**	**1.47**	**0.14**
3		**151.39**	**0.25**	**151.39**	0.45	**151.39**	0.31	**4.94**	**0.18**	**4.94**	0.20	**4.94**	0.21
1		**61.41**	1.02	**61.41**	1.51	**61.41**	**0.95**	**1.44**	**0.42**	**1.44**	**0.42**	**1.44**	0.48
2	10	**133.89**	0.89	**133.89**	**0.83**	**133.89**	1.03	**4.16**	0.64	**4.16**	**0.40**	**4.16**	0.86
3		**96.30**	**0.82**	**96.30**	1.08	**96.30**	0.96	**2.33**	**0.49**	**2.33**	0.50	**2.33**	1.05
1		**107.13**	**2.68**	**107.13**	3.94	**107.13**	4.22	**3.18**	1.26	**3.18**	**1.18**	**3.18**	1.53
2	15	**168.53**	**3.71**	**168.53**	3.94	**168.53**	4.14	**5.22**	2.32	**5.22**	2.66	**5.22**	**2.31**
3		**89.38**	**2.87**	**89.38**	3.80	**89.38**	3.44	**2.20**	1.18	**2.20**	**1.14**	**2.20**	1.73
1		199.54	**158.61**	245.92	179.83	**193.21**	265.00	**4.95**	88.38	**4.95**	**84.90**	**4.95**	122.73
2	50	215.91	190.19	210.79	**142.97**	**201.93**	259.05	**4.98**	88.37	**4.98**	**87.94**	**4.98**	88.51
3		319.36	**198.94**	309.61	205.34	**206.97**	219.24	**4.70**	78.99	**4.70**	**76.20**	**4.70**	95.02
1		361.27	694.67	300.67	**691.61**	**262.19**	1087.91	**4.65**	258.49	**4.65**	**241.76**	**4.65**	389.62
2	75	352.86	894.16	345.48	**695.86**	**303.93**	929.34	**4.75**	270.75	**4.75**	**256.27**	**4.75**	330.12
3		356.88	642.35	345.92	**616.19**	**313.82**	1120.79	**5.31**	330.51	**5.31**	**316.30**	**5.31**	397.31
1		**854.00**	**98.01**	**854.00**	100.31	**854.00**	100.33	**9.58**	**97.44**	**9.58**	107.84	**9.58**	104.43
2	100	519.33	**1730.32**	511.40	2193.40	**397.67**	3420.88	**5.05**	591.96	**5.05**	**504.19**	**5.05**	739.52
3		402.24	2164.14	394.88	**1849.67**	**385.18**	2378.72	**4.83**	**552.17**	**4.83**	596.69	**4.83**	928.85
Avg		254.39	376.89	250.90	**371.73**	**228.69**	544.27	**4.33**	131.33	**4.33**	**126.60**	**4.33**	178.04

distribution metrics can be found in Supplementary Material E. Table 8 presents the average values of each instance size (n) for the best metaheuristic from Table 7. In the columns, we provide the customers' instance size (n), the operational costs objective function (ω) in euros (€), the makespan objective function (τ) in hours (h), vehicles' costs in euros (€), distance in kilometers (km), the number of customers served by each transportation mode (#), i.e., trucks, drones, and robots, the number of customers served by drone multi-legs (#ML), and the number of customers served by drone round-trips (#RT). The information corresponding to trucks, drones, and robots is shown by the letters T, D, and R, respectively. We present in boldface the best objective values and computational times. Additionally, Fig. 6 shows the behavior of ω and τ when minimizing operational costs (a) and the makespan (b).

In Supplementary Material E, we present the results of the non-parametric Friedman test used to assess significant performance differences among the metaheuristics, followed by the Nemenyi post-hoc test for pairwise comparisons. The results show statistically significant differences in the performance of the metaheuristics, with p-values of 0.000006 and 0.000009 for the operational cost and makespan objectives, respectively, both well below the $p < 0.05$ threshold. As summarized in Fig. 5, the results confirm that 3P-GMS-ILS consistently achieves the lowest normalized objective values and outperforms the other metaheuristics, exhibiting the lowest mean, standard deviation, standard error of the mean, and the narrowest confidence intervals, with values of 1.0329, 0.0688, 0.0162 and [1.9987, 1.0671] for the operational costs objective, and values of 1.0, 0.0, 0.0, and [0.0, 0.0] for the makespan objective, respectively.

The results from Table 7 show that the best metaheuristic approach for the small and large set of instances of Amsterdam is the 3P-GMS-ILS. The 3P-GMS-ILS outperforms the GVNS and SA with average objective values of 228.69 and 4.33 for the operational costs and makespan objectives, respectively. In addition, we observe that GVNS, SA, and 3P-GMS-ILS yield identical results for Instance #1 of 100 customers across both the operational-costs and makespan objectives. This outcome can be attributed to the fact that all three metaheuristics share the same set of neighborhood operators and employ the same constructive heuristic for generating initial solutions (i.e., Phase I). Consequently, despite their differing exploration strategies, they converge to the same local optimum, highlighting the complexity and challenging landscape of the studied instance. In addition, SA is the most efficient metaheuristic for the operational costs objective, with average computational times of 500.51 [s], whereas 3P-GMS-ILS presents the lowest computational times for the makespan objective with average values of 178.04 [s]. This shows that the 3P-GMS-ILS is an effective metaheuristic approach that, by combining a multi-start mechanism with granular neighborhoods, is able to outperform a state-of-the-art GVNS for the VRPD-RS with the lowest computational times for the makespan objective. Furthermore, our results correlate with the results in Morim et al. (2024), where the GVNS outperforms the SA metaheuristic for the VRPD-RS.

The results in Table 8 show that ω achieves the best results when minimizing operational costs with average values of 228.69, whereas τ shows the best results when minimizing the makespan with average values of 4.33. It is worth highlighting that if we



(a) Performance for the operational-costs objective (b) Performance for the makespan objective

Fig. 5. Distribution of normalized objective values for GVNS, SA, and 3P-GMS-ILS.

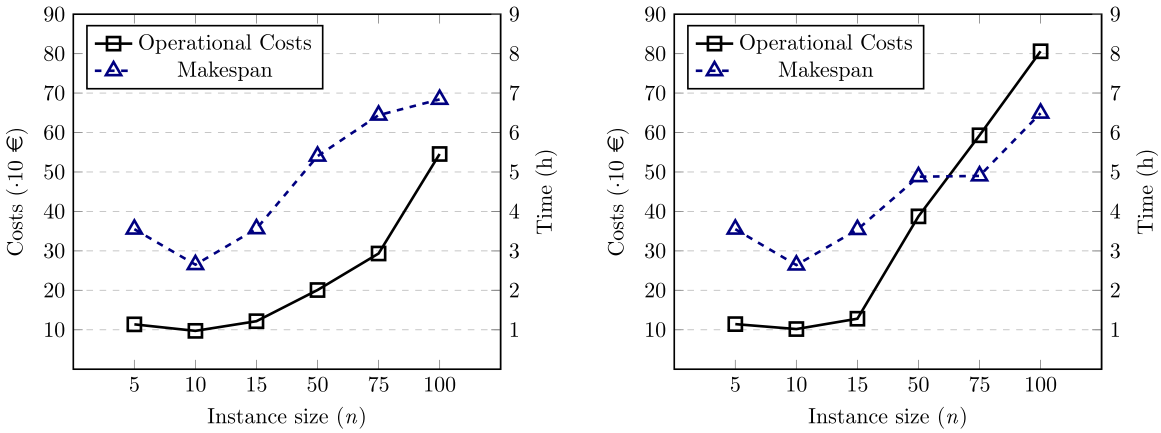
Table 7
Comparison of metaheuristic approaches for the operational costs and makespan objective functions.

Instance		Operational Costs						Makespan					
		GVNS		SA		3P-GMS-ILS		GVNS		SA		3P-GMS-ILS	
#	n	ω	Time [s]	ω	Time [s]	ω	Time [s]	τ	Time [s]	τ	Time [s]	τ	Time [s]
1		135.47	**0.21**	**132.23**	7.65	**132.23**	0.32	4.45	**0.05**	**4.24**	7.31	**4.24**	0.25
2	5	131.62	**0.06**	62.63	7.25	**57.35**	0.25	4.19	**0.04**	**1.47**	7.42	**1.47**	0.14
3		151.53	**0.06**	**151.39**	6.96	**151.39**	0.31	**4.94**	**0.05**	**4.94**	7.20	**4.94**	0.21
1		159.20	**0.80**	65.95	11.79	**61.41**	0.95	5.11	**0.47**	**1.44**	10.44	**1.44**	0.48
2	10	147.53	**0.86**	138.04	11.95	**133.89**	1.03	4.68	**0.37**	**4.16**	12.00	**4.16**	0.86
3		157.01	5.24	105.23	12.72	**96.30**	**0.96**	5.10	**0.45**	4.98	11.49	**2.33**	1.05
1		161.73	5.49	116.36	17.48	**107.13**	**4.22**	5.20	2.63	**3.18**	12.28	**3.18**	**1.53**
2	15	168.96	4.76	169.46	16.51	**168.53**	**4.14**	5.31	2.59	**5.22**	16.87	**5.22**	**2.31**
3		146.00	4.85	123.74	16.07	**89.38**	**3.44**	4.71	2.29	**2.20**	16.02	**2.20**	**1.73**
1		209.71	773.52	245.03	**152.67**	**193.21**	265.00	**4.95**	3600.43	**4.95**	148.46	**4.95**	**122.73**
2	50	240.11	3601.62	331.39	**113.43**	**201.93**	259.05	5.04	3600.37	5.04	132.10	**4.98**	**88.51**
3		317.32	3601.70	313.84	**185.02**	**206.97**	219.24	4.93	3601.05	4.84	239.66	**4.70**	**95.02**
1		385.70	3600.55	343.34	**255.42**	**262.19**	1087.91	5.26	3601.70	5.18	**240.11**	**4.65**	389.62
2	75	366.86	3602.17	433.75	**223.66**	**303.93**	929.34	5.17	3601.13	4.77	**247.70**	**4.75**	330.12
3		369.44	3604.96	576.79	**192.28**	**313.82**	1120.79	**5.31**	3602.44	5.45	**205.31**	**5.31**	397.31
1		**854.00**	3608.17	**854.00**	3600.81	**854.00**	**100.33**	**9.58**	3603.75	**9.58**	3604.79	**9.58**	**104.43**
2	100	967.63	3609.56	967.63	3605.26	**397.67**	**3420.88**	8.96	3605.33	8.96	3601.76	**5.05**	**739.52**
3		423.95	3617.28	589.32	**572.255**	**385.18**	2378.72	5.41	3620.80	5.66	**620.43**	**4.83**	928.85
Avg		305.21	1646.77	317.78	**500.51**	**228.69**	544.27	5.46	1802.55	4.79	507.85	**4.33**	**178.04**

compare how the values of ω and τ vary for both objective functions, the difference in τ, i.e., 4.33 and 4.74, is significantly smaller than the difference in ω, i.e, 228.69 and 355.11. This shows that minimizing the makespan disregards operational costs, whereas minimizing operational costs reduces both operational costs and the makespan. Moreover, trucks show the highest costs in the delivery system, whereas drones and robots present the lowest costs for both objective functions. These costs correlate with the vehicles' utilization, where trucks, drones, and robots serve, on average, 36.22, 1.72, and 4.56 customers for the operational costs objective function, respectively, whereas they serve 31.00, 4.00, and 7.50 customers for the makespan objective function, respectively. Therefore, we conclude that for scenarios where trucks have lower routing costs than drones and robots, the VRPD-RS-TW serves more customers by trucks. In particular, the low utilization of drones can be explained by the synchronization of the truck and drone deliveries. For instance, in case a truck arrives first at the meeting point, the truck should wait for the drone, generating higher operational costs due to the drivers' wages. Consequently, limited simultaneous coordination of deliveries may reduce the routing costs of drones and robots, but can significantly increase overall operational costs due to driver wages incurred while trucks wait for the return of drones and robots.

Table 8
Comparison of the operational costs and makespan objective functions for the 3P-GMS-ILS.

Instance		Operational Costs												
Set	n	ω	τ	T [€]	D [€]	R [€]	T [km]	D [km]	R [km]	#T	#D	#R	#ML	#RT
Small Ring	5	113.66	3.55	108.49	2.66	2.51	7.98	2.66	2.51	2.00	1.67	1.33	1.00	0.67
	10	97.20	2.65	82.12	3.26	11.82	10.91	3.26	11.82	3.67	2.33	4.00	1.00	1.33
	15	121.68	3.56	112.23	1.00	8.45	22.13	1.00	8.45	10.00	1.33	3.67	0.67	0.67
Large Ring	50	200.70	5.40	188.93	6.90	4.87	107.19	6.90	4.87	44.67	2.33	3.00	2.33	0.00
	75	293.31	6.44	269.35	2.68	21.28	166.38	2.68	21.28	65.00	1.33	8.67	1.33	0.00
	100	545.62	6.84	529.20	1.78	14.63	326.38	1.78	14.63	92.00	1.33	6.67	1.33	0.00
Avg		**228.69**	4.74	215.05	3.05	10.59	106.83	3.05	10.59	36.22	1.72	4.56	1.28	0.44
Instance		Makespan												
Set	n	ω	τ	T [€]	D [€]	R [€]	T [km]	D [km]	R [km]	#T	#D	#R	#ML	#RT
Small Ring	5	114.28	3.55	109.39	2.38	2.51	11.60	2.38	2.51	2.67	1.00	1.33	1.00	0.00
	10	101.83	2.64	84.60	2.83	14.40	21.27	2.83	14.40	4.33	1.00	4.67	1.00	0.00
	15	128.02	3.54	113.74	2.40	11.89	30.57	2.40	11.89	8.67	1.67	4.67	0.33	1.33
Large Ring	50	387.51	4.88	349.34	13.92	24.25	233.74	13.92	24.25	40.33	3.00	6.67	2.00	1.00
	75	593.00	4.90	511.93	33.78	47.29	294.41	33.78	47.29	54.67	8.00	12.33	4.00	4.00
	100	806.03	6.49	682.72	60.68	62.63	413.96	60.68	62.63	75.33	9.33	15.33	4.67	4.67
Avg		355.11	**4.33**	308.62	19.33	27.16	167.59	19.33	27.16	31.00	4.00	7.50	2.17	1.83



(a) Operational Costs (b) Makespan

Fig. 6. Comparison of the operational costs and makespan objective functions for the VRPD-RS-TW.

5.2.4. Comparison of the VRPD-RS-TW with similar transportation systems

This set of experiments compares the VRPD-RS-TW with the VRP-TW, VRPD-TW, and VRP-RS-TW when solving the small and large sets of instances of Amsterdam and minimizing operational costs. Due to the size of the instances, we apply the 3P-GMS-ILS and consider the same setup as in Section 5.2.2. We adopt a non-uniform cost structure where drones and robots incur €1 per arc, reflecting maintenance, remote piloting, insurance, battery degradation, and depreciation. In contrast, trucks are charged a lower cost of €0.25 per arc, capturing economies of scale and established infrastructure. This generates more realistic scenarios, distinguishing them from the uniform costs settings presented in Section 5.2.5.

The results are listed in Table 9, which averages per instance size (n) the information presented in Supplementary Material F for the VRPD-RS-TW, VRP-TW, VRPD-TW, and VRP-RS-TW. In the columns, we provide the customers' instance size (n), the operational costs objective function (ω) in euros (€), the makespan objective function (τ) in hours (h), and the number of customers served by trucks (#T). Furthermore, we present in boldface the best values for the objective functions and the smallest values for the number of customers served by trucks. Furthermore, Fig. 7 presents the behavior of ω in the left-hand side (a) and τ in the right-hand side (b) when minimizing the operational costs objective function for the different transportation systems.

Results show the benefits of multimodality in the delivery system. Here, the VRPD-RS-TW presents the lowest operational costs and the VRP-TW shows the highest operational costs. These costs correlate with the utilization of trucks (#T), where the VRPD-RS-TW provides the lowest utilization of trucks, i.e., the highest simultaneous coordination of deliveries. Therefore, even though the routing costs of drones and robots are higher than trucks' routing costs, the VRPD-RS-TW delivers more parcels by robots and drones than the VRP-TW, VRPD-TW, and VRP-RS-TW, outperforming the objective function values (ω) of these related transportation systems.

Table 9
Comparison of the operational costs and makespan objective functions for the small and large ring of Amsterdam, when minimizing the operational costs objective function.

Instance		VRP-TW			VRPD-TW			VRP-RS-TW			VRPD-RS-TW		
Set	n	ω	τ	#T	ω	τ	#T	ω	τ	#T	ω	τ	#T
Small Ring	5	144.93	4.62	5.00	138.40	4.46	3.00	122.90	3.80	3.67	**113.66**	**3.55**	2.00
	10	157.68	5.00	10.00	152.46	4.87	7.00	104.95	2.84	5.67	**97.20**	**2.65**	3.67
	15	160.08	5.08	15.00	158.88	5.05	13.33	126.43	3.67	11.00	**121.68**	**3.56**	10.00
Large Ring	50	209.52	5.97	50.00	239.27	5.49	48.67	201.13	**5.33**	44.33	**200.70**	5.40	44.67
	75	338.13	6.49	75.00	344.61	**6.43**	72.00	305.76	6.45	66.00	**293.31**	6.44	65.00
	100	743.02	8.24	100.00	557.12	7.32	97.00	735.64	7.99	97.67	**545.62**	**6.84**	92.00
Avg		292.23	5.90	42.50	265.12	5.60	40.17	266.14	5.02	38.06	**228.69**	**4.74**	36.22

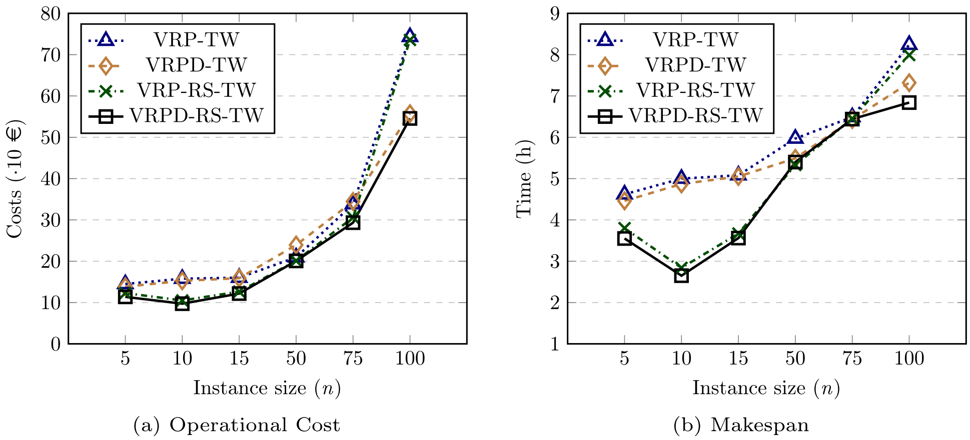


Fig. 7. Comparison of operational costs and makespan, when minimizing the operational costs objective function.

We emphasize that the VRPD-TW, VRP-RS-TW, and VRPD-RS-TW face energy constraints due to the battery of drones and robots. Furthermore, drones and robots execute different types of delivery operations and interactions with trucks. For robot deliveries, the truck delivers the parcels to a robot station and then leaves. For drone deliveries, trucks might have to wait for drones at meeting points. Thus, deploying drones to simultaneously execute deliveries could also cause higher operational costs due to drivers' wages, resulting in costly deliveries. Nevertheless, results show that by implementing more than one autonomous modality of transport, i.e., drones and robots, the VRPD-RS-TW simultaneously coordinates more delivery operations than the VRPD-TW and VRP-RS-TW, achieving lower operational costs when facing time-window and energy constraints.

5.2.5. Sensitivity analysis of the transportation costs for the VRPD-RS-TW and similar transportation systems

In this set of experiments, we consider the transportation costs of drones and robots to be the same as the truck costs for the VRP-TW, VRPD-TW, VRP-RS-TW, and VRPD-RS-TW, i.e., $c^T_{ij} = c^D_{ij} = c^R_{ij} = 0.25\ \forall\ (i,j) \in A$. We adopt this cost structure to eliminate potential biases associated with the configuration used in Section 5.2.4 and to assess whether the observed trends hold in scenarios where trucks, drones, and robots incur identical routing costs, e.g., due to economies of scale. Results are presented in Table 10, which lists the same columns as Table 9 and reports the average values of each instance size (n) summarizing the information presented in Supplementary Material F for the VRP-TW and Supplementary Material G for the VRPD-TW, VRP-RS-TW, and VRPD-RS-TW. Note that the results for the VRP-TW are extracted from Table 9 since the VRP-TW is an only-truck transportation system whose transportation costs are not modified. The best values for the objective functions and the smallest values for the number of customers served by trucks are boldfaced. Furthermore, Fig. 8 illustrates the behavior of ω in the left-hand side (a) and τ in the right-hand side (b) when minimizing the operational costs objective function for the different transportation systems.

Results show the same tendency as the experiments in Section 5.2.4, where the VRPD-RS-TW is the delivery system with the lowest operational costs and the highest simultaneous coordination of deliveries, i.e., lowest values of #T. In particular, if we compare the results of the VRPD-RS-TW in Tables 9 and 10, as expected, by decreasing the routing costs of drones and robots, the VRPD-RS-TW increases the number of drone and robot deliveries. As a result, this higher simultaneous coordination of deliveries generates lower operational costs and makespan values. Consequently, we conclude that, for the instances of Amsterdam, when considering the same routing costs for trucks, drones, and robots, the VRPD-RS-TW outperforms the related transportation systems, as well as increases the share of drone and robot deliveries in comparison to the scenarios where robots and drones present higher routing costs than trucks.

Table 10
Comparison of the operational costs and makespan objective functions for the small and large ring of Amsterdam, when minimizing the operational costs objective function with the same routing costs for trucks, drones, and robots.

Instance		VRP-TW			VRPD-TW			VRP-RS-TW			VRPD-RS-TW		
Set	n	ω	τ	#T	ω	τ	#T	ω	τ	#T	ω	τ	#T
Small Ring	5	144.93	4.62	5.00	136.39	4.46	2.67	120.98	3.80	3.00	**109.75**	**3.55**	1.67
	10	157.68	5.00	10.00	149.55	4.87	4.67	95.56	2.84	5.00	**85.84**	**2.65**	3.33
	15	160.08	5.08	15.00	157.06	5.03	9.67	119.57	3.64	10.00	**113.62**	**3.54**	6.00
Large Ring	50	209.52	5.97	50.00	193.70	5.35	40.33	180.63	4.89	37.67	**179.12**	**4.88**	33.33
	75	338.13	6.49	75.00	336.99	6.29	61.67	203.67	5.08	45.00	**202.71**	**5.05**	39.67
	100	743.02	8.24	100.00	381.75	5.75	82.67	685.42	8.01	84.33	**233.81**	**5.09**	46.67
Avg		292.23	5.90	42.50	225.91	5.29	33.61	234.31	4.71	30.83	**154.14**	**4.12**	21.78

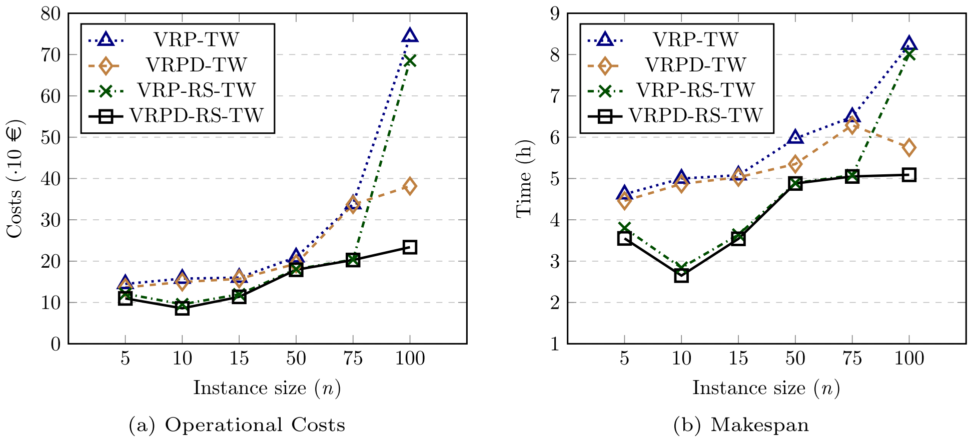


Fig. 8. Comparison of transportation systems, when minimizing the operational costs, with the same routing costs for trucks, drones, and robots.

5.2.6. Sensitivity analysis on the 3P-GMS-ILS parameters for the operational-costs objective function

This section presents the results of a sensitivity analysis to evaluate how variations in the 3P-GMS-ILS parameters impact the performance of the metaheuristic algorithm for the VRPD-RS-TW. The focus is primarily on operational costs, as they represent a more challenging objective function to optimize. For a sensitivity analysis on the makespan objective function, please refer to Supplementary Material H. As detailed in Section 5.1.1, we applied the Friedman non-parametric statistical test to assess the performance of the 3P-GMS-ILS for the VRPD-RS-TW, identifying $R = 25$ and $\beta = 3.0$ as the default values for the maximum number of multi-start procedures and the sparsification parameter, respectively.

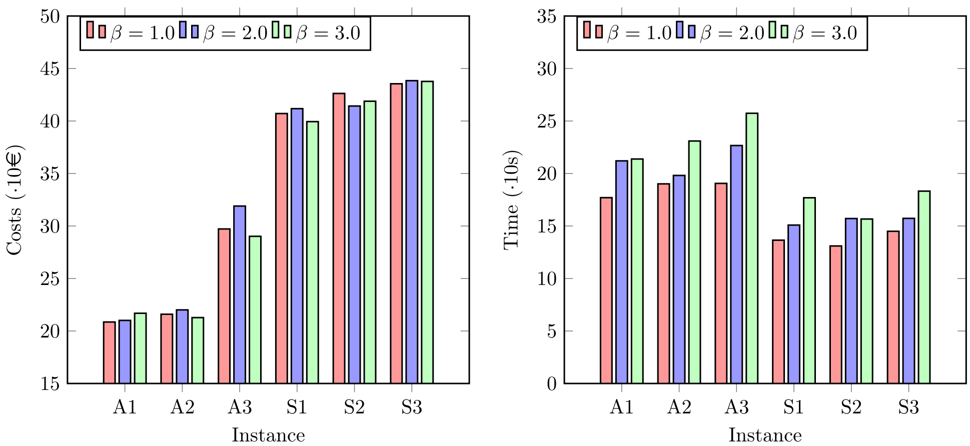
The experiments were executed using Amsterdam-based instances of 50 customers, selected due to their representativeness of the entire dataset and their manageable computational times compared to instances of 75 and 100 customers. Additionally, to explore different scenario settings, we generated three new instances of 50 customers, with uniformly distributed customer locations, robot stations, and the depot, within a square city of 20x20 [km] of area. In the experiments, we tested 6 parameter combinations, across 6 instances, with two objective functions, and ran 10 executions per instance. This resulted in a total of $6 \cdot 6 \cdot 2 \cdot 10 = 720$ executions of the 3P-GMS-ILS. Table 11 presents the average objective values and computational times of the 3P-GMS-ILS, when minimizing operational costs (ω) for combinations of $R = 15$ and $\beta = \{1.0, 2.0, 3.0\}$. Fig. 9 summarizes the results of Table 11 for the objective function values and computational times. Table 12 presents the average objective values and computational times of the 3P-GMS-ILS, when minimizing operational costs for combinations of $R = \{5, 15, 25\}$ and $\beta = 3.0$. Fig. 10 summarizes the results of Table 12 for the objective function values and computational times.

The results in Table 11 indicate that the average objective values generally decrease as the sparsification parameter β increases from 1.0 to 3.0. This behavior is consistent with the expectations outlined in Eq. (46), where a larger β leads to a higher acceptance threshold ϑ. Consequently, as ϑ increases, more arcs are included in the neighborhood set $\mathcal{N}_l(\cdot)$, allowing the algorithm to explore a wider set of candidate solutions. However, this trend does not always hold. For instance, the average objective values for $\beta = 2.0$ are higher than those for $\beta = 1.0$. This can be attributed to the fact that, while larger β values expand the neighborhood and promote greater exploration, they also increase the likelihood of accepting solutions that include poor-quality arcs. As a result, although higher β values tend to yield better objective values overall, they also pose a risk of deteriorating solution quality in specific cases, as illustrated by the outcome for $\beta = 2.0$. Furthermore, as shown in Fig. 9, the advantage of having large β values comes with a trade-off, where higher values of the sparsification parameter correspond to increased computational times.

As a result, although increasing the sparsification parameter β improves solution quality by expanding the search space, it also leads to longer computational times. Therefore, we conclude that the sparsification parameter β serves as an effective mechanism

Table 11
Analysis of 3P-GMS-ILS for the operational-costs objective function when varying the sparsification parameter β.

Instance			$R = 25,\ \beta = 1.0$		$R = 25,\ B = 2.0$		$R = 25,\ \beta = 3.0$	
Set	#	n	ω	Time [s]	ω	Time [s]	ω	Time [s]
Amsterdam	1	50	**208.57**	**176.99**	210.09	212.03	216.90	213.80
	2		215.97	**190.12**	220.10	198.15	**212.76**	230.95
	3		297.18	**190.58**	319.00	226.64	**290.17**	257.36
Square	1	50	407.11	**136.51**	411.76	150.88	**399.41**	176.88
	2		426.25	**130.98**	**414.28**	157.10	418.85	156.64
	3		**435.46**	**144.97**	438.42	157.27	437.70	183.22
Avg			331.76	**161.69**	335.61	183.68	**329.30**	203.14



(a) Operational Costs (b) Computational Times

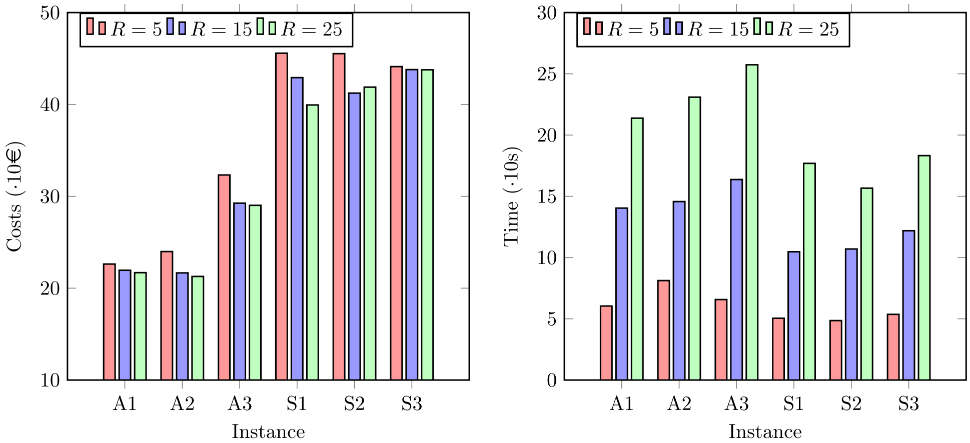
Fig. 9. Comparison of operational-costs and computational times when $R = 25$ and $\beta = \{1.0, 2.0, 3.0\}$ for the 3P-GMS-ILS.

Table 12
Analysis of 3P-GMS-ILS for the operational-costs objective function when varying the number of multi-start procedures R.

Instance			$R = 5,\ \beta = 3.0$		$R = 15,\ \beta = 3.0$		$R = 25,\ \beta = 3.0$	
Set	#	n	ω	Time [s]	ω	Time [s]	ω	Time [s]
Amsterdam	1	50	226.26	**60.40**	219.51	140.37	**216.90**	213.80
	2		239.86	**81.21**	216.61	145.70	**212.76**	230.95
	3		323.21	**65.71**	292.55	163.69	**290.17**	257.36
Square	1	50	455.77	**50.44**	429.18	104.73	**399.41**	176.88
	2		455.28	**48.55**	**412.33**	106.98	418.85	156.64
	3		441.18	**53.63**	437.94	121.90	**437.70**	183.22
Avg			356.93	**59.99**	334.69	130.56	**329.30**	203.14

within the granular neighborhoods of the 3P-GMS-ILS, enabling more efficient exploration of high-potential areas of the solution space. Consequently, it is crucial to establish a proper balance between the quality of the solutions and the maintenance of computational efficiency when selecting the sparsification parameter β in the 3P-GMS-ILS.

The results shown in Table 12 indicate that the average objective values decrease as the maximum number of multi-start procedures R increases from 5 to 15 and 25. This trend is consistent with the metaheuristic paradigm of the 3P-GMS-ILS, where a larger number of iterations enables a more extensive exploration of the solution space, increasing the likelihood of discovering better solutions. As R increases, the 3P-GMS-ILS algorithm achieves a greater refinement of its solutions through a more extensive application of neighborhood operators $\mathcal{N}_l(\cdot)$. Conversely, a limited number of iterations may cause the algorithm to converge prematurely to suboptimal or low-quality local optima. Thus, with an increased number of iterations, the algorithm can apply additional shaking procedures $\mathcal{SH}_k(\cdot)$, which enhances diversification and can guide the search towards promising areas of the solution space. However, as illustrated in Fig. 10, increasing the number of iterations also results in significantly higher computational times. In particular, when the number of iterations increases from 5 to 15 and 25, the solution quality improves by $\approx 6.23\,\%$ and $1.61\,\%$, respectively, while computational times increase by $\approx 117.64\,\%$ and $55.59\,\%$. These numbers are expected to grow further for larger instances of 75



(a) Operational Costs (b) Computational Times

Fig. 10. Comparison of operational-costs and computational times when $R = \{5, 15, 25\}$ and $\beta = 3.0$ for the 3P-GMS-ILS.

and 100 customers. Therefore, similar to the sparsification parameter β, it is essential to establish an appropriate balance between solution quality and computational efficiency when selecting the maximum number of iterations R in the 3P-GMS-ILS algorithm.

Additionally, the sensitivity analysis of the makespan objective function, presented in Supplementary Material H, is consistent with the findings for the operational-cost objective in terms of computational efficiency. Specifically, higher values of β result in increased computational times, as more arcs are explored within the neighborhood structures, reducing the efficiency of the metaheuristic. Similarly, increasing the value of R significantly raises computational time, highlighting its impact on performance. However, with respect to solution quality, most combinations of R and β yield similar objective values for the makespan, indicating a more stable behavior of the metaheuristic for this objective. This suggests that, unlike the operational-cost objective, where broader exploration improves results, the makespan objective can be effectively optimized with a less extensive search, still producing high-quality solutions.

6. Managerial insights and discussion

Section 6.1 provides an in-depth discussion of the assumptions underlying the VRPD-RS-TW and examines their impact on practical implementations, while Section 6.2 presents key insights derived from the experimental results.

6.1. Discussion

Integrating multiple transportation modes, namely trucks, drones, and sidewalk robots, into a unified last-mile delivery system presents considerable challenges in coordinating and synchronizing delivery activities. While our model assumes seamless cooperation among these modes, real-world operations are often disrupted by uncertainties such as weather conditions, traffic congestion, battery constraints, and variability in service times. These factors can significantly degrade system performance. Drones, in particular, are highly sensitive to energy availability, with unexpected depletion caused by adverse weather or inefficient energy usage, risking mid-route failures that compromise safety and delivery reliability. Additionally, delays in modal handoffs pose a major challenge. If trucks experience setbacks due to traffic or mechanical issues, subsequent transfers of parcels or batteries to drones or robots may be postponed, triggering cascading delays throughout the delivery schedule.

To mitigate such operational uncertainties, it is essential to incorporate reliable contingency strategies, such as real-time monitoring, predictive maintenance, and dynamic routing adjustments. In this context, Robust Optimization (RO) provides a powerful modeling framework to handle uncertainty by generating solutions that remain feasible and effective across a wide range of potential scenarios. Rather than depending on precise input data, RO models account for variability in key parameters, such as travel times, drone availability, or delivery time windows, by introducing uncertainty sets that represent possible deviations. This enables planners to proactively design solutions that perform well even under adverse conditions, enhancing the system's resilience and service reliability. Our proposed modeling framework and metaheuristic provide a strong foundation for further research in this direction. They can serve as a baseline for developing and testing robust optimization approaches that explicitly address real-world uncertainties in integrated last-mile logistics, advancing the design of more adaptive, dependable, and scalable delivery systems.

While our current model enforces synchronization between vehicles through strict timing constraints, we acknowledge that explicitly minimizing synchronization delays, such as waiting times between trucks, drones, and robots, could provide additional insights into the trade-offs between routing efficiency and coordination. Future work could explore this dimension by incorporating synchronization delays, or vehicles' idle times, directly into the objective function. Alternatively, delay penalties could be modeled dynamically based on service-level agreements or stochastic disruptions, enabling more flexible and realistic coordination strategies.

Such extensions would enhance the model's applicability to real-world settings where delays and imperfect timing are common and costly.

Incorporating regulatory and infrastructure constraints into metaheuristic optimization is crucial for ensuring realistic and compliant last-mile parcel delivery. In our current approach, drones are prevented from taking off or landing at nodes within no-fly zones, but we do not restrict flight paths that pass over these areas, which is an important limitation in regions with stricter regulations. To address this, we recommend integrating geospatial data into the optimization process, treating no-fly zones as restricted throughout the entire drone trajectory. This can be achieved by forbidding arcs that intersect no-fly zones or applying spatial clustering techniques to guide path generation. Such enhancements would bring the model closer to reality and support the development of robust and regulation-compliant multimodal last-mile delivery systems.

Furthermore, in the mathematical formulation, we omitted sidewalk-access constraints, as our primary experiments focus on Amsterdam, a city and country characterized by predominantly flat terrain and well-maintained cycling infrastructure. Consequently, we do not model slope or grade limits, since the Netherlands' elevation profile rarely challenges delivery robots' traction or stability. Additionally, we assume consistently smooth pavement by having robots operate on bike lanes, disregarding surface-quality assessments in transportation times and accessibility. We also exclude time-of-day restrictions on sidewalk or bike-lane use, e.g., peak-hour bans in crowded pedestrian zones. All these features are beyond our research scope, but we encourage their integration in future works, especially when considering other application areas.

6.2. Managerial insights

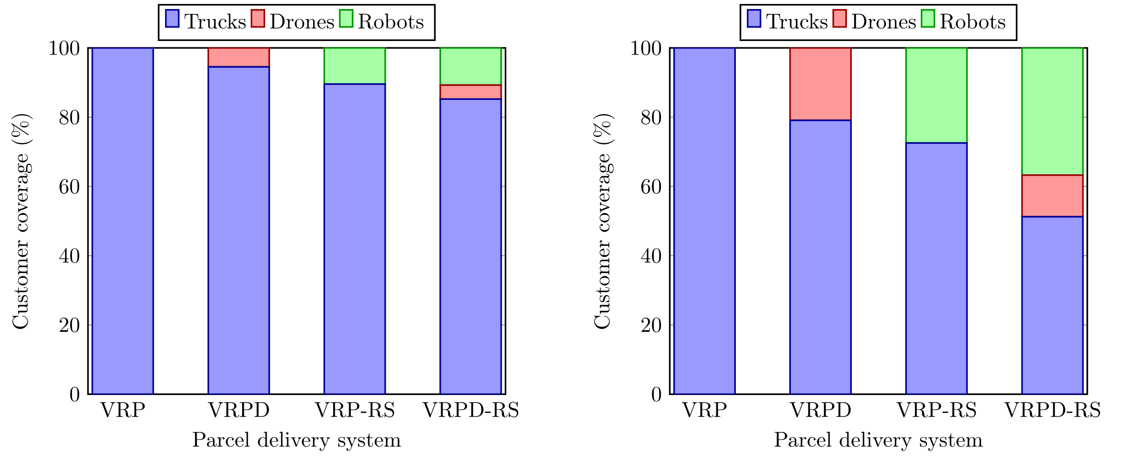
By incorporating drone hovering times, energy consumption constraints related to parcel weight, and time window constraints, the VRPD-RS-TW significantly extends the complexity of the problem studied in Morim et al. (2024). This increased complexity is evident in the results presented in Section 5.2.1, where the MILP model fails to reach optimal solutions for instances with more than 5 nodes. Moreover, for instances with 10 and 15 customers, the model shows optimality gaps exceeding 58 % when drivers' wages are incorporated into the objective function. This shows that for realistic implementations, instances larger than 50 nodes, exact optimization approaches are not suitable for solving the VRPD-RS-TW. Nevertheless, the MILP offers a strong foundation for mathematically defining autonomous multimodal delivery systems and serves as a rigorous benchmark for evaluating metaheuristic approaches designed to solve large-scale instances.

One of the core strengths of the 3P-GMS-ILS lies in its integration of the *multi-start* mechanism and *granular neighborhoods*. As demonstrated in Section 5.2.2, *granular neighborhoods* reduce computational time while enhancing the metaheuristic performance in minimizing operational costs. Similarly, the *multi-start* mechanism contributes to improved objective values, however, at the expense of increased computational time due to repeated re-initializations of the search process. As shown in Section 5.2.3, 3P-GMS-ILS consistently outperforms both GVNS and SA across the two studied objective functions, validating the effectiveness of its design components. Nevertheless, there remains room for improvement through further research, as we do not observe significant differences when solving the makespan objective in Section 5.2.2. In particular, we suggest exploring alternative criteria to trigger the *multi-start* procedures and refine the use of *granular neighborhoods*. Adopting diverse multi-start strategies that do not exclusively depend on incumbent solutions may promote broader exploration of the solution space, potentially uncovering high-quality, previously unexplored regions. Additionally, performance could be enhanced by incorporating novel *granular neighborhood* strategies. Promising directions include the use of *adaptive granular neighborhoods*, which dynamically adjust granularity based on search progress, and *granular neighborhood decomposition*, which leverages problem structure to guide local search more efficiently.

In Sections 5.2.4 and 5.2.5, we compare the VRP-TW, VRPD-TW, VRP-RS-TW, and VRPD-RS-TW across two cost settings for the operational cost objective. Fig. 11 displays the percentage of customer coverage achieved by each transport mode in both experimental settings. For brevity, the acronym for time windows (TW) is omitted in the problem descriptions shown in Fig. 11. In the results, the VRPD-RS-TW consistently outperforms the VRP-TW, generating objective value improvements of 21.74 % and 47.25 % in Sections 5.2.4 and 5.2.5, respectively. These improvements correspond to drone-and-robot customer coverages of 14.78 % in Section 5.2.4 and 48.75 % in Section 5.2.5. This indicates that, under the cost setting of Section 5.2.4, even a modest incorporation of drones and robots can lead to substantial reductions in objective values. It is worth noticing that this limited participation is not a result of problem constraints or inadequate exploration of the solution space, but rather reflects the goal of obtaining better objective values. Consequently, under the cost setting of Section 5.2.5, drones and robots assume a much larger role in customer coverage for the same instances. This demonstrates that in future scenarios, where the costs of drones and robots are expected to decrease due to economies of scale, autonomous multimodal delivery systems could increasingly benefit from the coordinated use of both modes.

Moreover, the drone-and-robot customer coverages of 14.78 % in Section 5.2.4 and 48.75 % in Section 5.2.5 may be potentially restricted, since we impose both hard time-window constraints and strict no-fly zone constraints on both drone and robot delivery operations. Without time-window constraints, drones could service longer-range requests that are otherwise infeasible under tight time windows. This is especially critical for truck-drone synchronization, where if trucks wait long times for drones, this may generate infeasible arrival times at the truck-served delivery locations. Similarly, robots could traverse extended routes to further reduce makespan and operational costs. Additionally, forbidding drone take-off, service, and landing within no-fly zones removes potential clusters of customers from its reachable set. Therefore, by relaxing these two sets of constraints, enabling flexible delivery intervals and permitting drone operations in currently restricted areas, the percentage customer coverage of drones and robots would increase in the scenarios studied in Sections 5.2.4 and 5.2.5.

Section 5.2.6 presents a sensitivity analysis of the two most critical parameters in the 3P-GMS-ILS: the number of multi-start procedures (R) and the sparsification parameter (β). These parameters not only influence computational efficiency but also shape the



(a) Distribution of vehicles' deliveries in Section 5.2.4. (b) Distribution of vehicles' deliveries in Section 5.2.5.

Fig. 11. Distribution of vehicles' metrics across different cost scenarios.

exploration of the solution space and the resulting solution quality. As anticipated, the analysis reveals that higher values of R and β lead to increased computational effort, as the metaheuristic converges to a greater number of new solutions and evaluates a larger set of arcs. In contrast, while larger values of R and β produce a noticeable improvement in the operational-costs objective, they have little to no significant impact on the makespan objective. Hence, compared to the operational-costs objective, less computational effort is needed for solving the makespan. On the other hand, for the studied combinations of R and β across both objectives, the computational efficiency of 3P-GMS-ILS ranges between 29.09 and 257.36 seconds for instances of 50 customers. Although the 3P-GMS-ILS is designed to support last-mile route planning decisions, its computation times of only a few minutes are acceptable for this application. However, for optimization systems that require real-time responsiveness to road disruptions, these computation times are too long. Therefore, we recommend further optimizing the 3P-GMS-ILS to deliver high-quality solutions with reduced computational effort for such time-sensitive applications.

Finally, in Supplementary Material I, we introduce a small set of experiments comparing the performance of the 3P-GMS-ILS when including and excluding drone energy-hovering constraints. The results show differences for the operational costs objective, while the same solutions are found for the makespan objective. This is in alignment with the results found in the previous sets of experiments, where no significant differences were found for the makespan objective. On the other hand, the operational costs objective shows the same results when including and excluding energy-hovering constraints for instances of 5 to 15 customers, whereas considerable differences were found in the routing schedules for instances of 50 customers. When excluding energy-hovering constraints, 3P-GMS-ILS finds better solutions than when considering hovering constraints. In instance 50-1, the solution of 3P-GMS-ILS becomes infeasible under hovering constraints, whereas the solutions for instances 50-2 and 50-3 remain feasible when assessing them under energy-hovering constraints. These results demonstrate the importance of the hovering constraints, which can generate infeasible solutions for instances of 50 customers. Furthermore, temporarily dropping energy-hovering constraints can lead the 3P-GMS-ILS metaheuristic to find better feasible solutions. This suggests a novel heuristic strategy for future work. Nevertheless, any final deployment must enforce hovering constraints to prevent drones from lingering idly on the ground, where they risk unexpected damage, rather than maintaining continuous hovering flight. Therefore, our results underscore the critical importance of enforcing energy-hovering constraints in large-scale instances.

7. Conclusions

This paper addresses the Drone-Assisted Vehicle Routing Problem with Robot Stations and Time Windows (VRPD-RS-TW). The VRPD-RS-TW incorporates multimodality into the delivery system to simultaneously coordinate delivery operations while minimizing operational costs.

To solve the VRPD-RS-TW, we presented a MILP formulation and developed a metaheuristic algorithm denoted by Three-Phased Granular Multi-Start Iterated Local Search (3P-GMS-ILS). To test the optimization approaches, we introduced a new set of instances of the city of Amsterdam, the Netherlands, and conducted five sets of experiments. We showed that the complexity of the problem substantially increases when including the drivers' wages in the operational costs objective function. The MILP formulation was only able to optimally solve instances of 5 customers when including the drivers' wages, whereas optimal solutions for instances of up to 15 customers were found when excluding the drivers' wages. Using more realistically sized instances, the 3P-GMS-ILS was not only able to outperform the MILP model but also a SA heuristic and a state-of-the-art GVNS (Morim et al., 2024). In addition, we demonstrated that the operational costs objective function not only reduces the total costs but also entails minimizing the makespan. Lastly, we found that the VRPD-RS-TW was able to outperform the VRP-TW, VRPD-TW, and VRP-RS-TW for the instance set of Amsterdam, with better average values for the operational costs and makespan. This conclusion stands for scenarios where (*i*) drone and robot routing costs are higher than truck routing costs, and (*ii*) truck, drone, and robot routing costs are the same. In particular, lower values in operational costs correlate with higher numbers of drone and robot deliveries, demonstrating that the higher the simultaneous coordination of deliveries, the lower the operational costs. Moreover, we found that drone and robot deliveries increased for scenarios where the routing costs are equal for the three modes of transport. These results are particularly interesting since, by deploying drones, trucks might have to wait for drones at the meeting point, generating higher costs due to drivers' wages. Therefore, we concluded that the incorporation of multimodality in the last mile creates more versatile delivery systems that can better handle the time-window and energy constraints to achieve lower operational costs than related transportation systems with fewer transport modalities. The final experiment focuses on a sensitivity analysis of the operational-costs objective function. For a representative set of instances, the results demonstrated that the sparsification parameter β effectively filters out poor-quality arcs, facilitating a more efficient exploration of the solution space. Furthermore, the results indicated that while an increase in the number of iterations leads to enhanced solutions, it also results in higher computational times.

This work provides a foundation for two main research directions. To outperform the proposed optimization methods, developing matheuristic algorithms to split the problem into sub-problems, which can be faced with exact and approximate methods, is a field of future research. Similarly, there is significant potential in exploring the performance of machine-learning enhanced heuristics in multimodal delivery systems. Integrating machine learning into the optimization process can more effectively guide the search towards promising regions of the solution space, while reducing computation times by avoiding searching less promising areas. Specifically, reinforcement learning (RL) holds great potential for addressing re-optimization challenges in last-mile disruption management. RL algorithms can learn to identify patterns and provide rapid and adaptive solutions in scenarios where delivery trucks encounter unexpected disruptions on the road and require immediate re-routing or rescheduling. On the other hand, last-mile delivery is characterized by multiple, interacting sources of uncertainty, for instance, uncertain arrival of parcels, uncertain flying times due to weather conditions, and traffic congestion. RL naturally handles these conditions by training across diverse simulated scenarios and producing adaptive policies rather than fixed routes, enabling real-time responsiveness. Future work would focus on risk-sensitive RL to extend its adaptability with formal guarantees against worst-case scenarios. These approaches can generate flexible, reliable delivery strategies that maintain high service quality and foster greater customer trust. Furthermore, this research provides insights into the question regarding the benefits of autonomous vehicles in last-mile delivery. Nevertheless, several other combinations of autonomous vehicles were not explored, such as robots serving trucks and drones deployed from stations. Future research could extend our investigation by (*i*) incorporating additional forms of multimodality and (*ii*) studying the impact of these delivery systems on other objective functions, such as latency and CO_2 emission. Finally, evaluating the studied transportation systems on a wider range of urban environments beyond the Amsterdam case and synthetic instances would provide valuable insights into their adaptability across diverse urban contexts.

CRediT authorship contribution statement

Giovanni Campuzano: Writing – original draft, Validation, Software, Methodology, Investigation, Funding acquisition, Formal analysis; **Eduardo Lalla-Ruiz:** Writing – review & editing, Supervision; **Martijn Mes:** Writing – review & editing, Project administration, Formal analysis.

Declaration of competing interest

The authors declare the following financial interests/personal relationships which may be considered as potential competing interests: Giovanni Campuzano reports financial support was provided by Chilean National Agency for Research and Development. If there are other authors, they declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Supplementary material

Supplementary material associated with this article can be found in the online version at 10.1016/j.tre.2025.104427

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